MANGALORE REFINERY & PETROCHEMICALS LTD., MANGALORE ,INDIA.

**Moodapadav, Kuthethoor P.O., via-
Katipalla,
Mangalore–575 030
India.**

TENDER for SUPPLY OF PLENUM CHAMBER & OTHER PARTS OF FRACTIONATOR SECTION FIN FANS IN HCU2 FOR REPLACEMENT OF DAMAGED SECTIONS

Supply of Plenum Chamber & other parts of Fractionator section Fin fans in HCU2 for replacement of damaged sections.

1.0 JOB DESCRIPTION:

It is envisaged to replace the following corroded parts / sections (as marked in the tender drawings) of the existing Fractionator sectionFinfans**EC42101** of the MRPL Refinery's HCU2 unit at Mangaluru:

1. Plenum Chamber with truss assembly.
2. Bottom cover plate with supporting structural members,
3. Fan Drive/motor suspension structural beam
4. Fan wire mesh guard.

Total number of Fin fans envisaged for replacement scope is **06nos**

Subject tender scope is to supply fabricated structural members / parts as marked in the tender drawings & any other structures /parts as per agreed scope,Correction / rectification works of the supplied items as per observations at Site during installation.

The Replacement / Installation of the supplied parts is in MRPL's Scope

Scope involves:

a) Procurement of raw materials/bought out items
b) Fabrication(at vendor's work site)
c) Painting(at vendor's work site)
d) Delivery /shifting of fabricated items to MRPL site.
e) Correction / rectification works of the supplied items as per observations at MRPL Site during installation(If not matching the tender drawings / approved drgs)

2.0 DETAILED SCOPE OF SUPPLY/WORK OF CONTRACTOR:

2.1 Replacement scope consists of the following parts / sections as marked in the tender drgs

1) Drawing No. 189811000 rev 2 for ec42101.
2) EC42101 three dimensional sketch of the fin fan cooler bottom duct.
3) BOM details of EC42101.

a) Plenum Chamber with truss assembly (IS 2062 std, plate 3.15mm T with G.I coated IS2629 std)

b) Bottom cover plate with supporting structural members , (plate IS 2062 std, plate 3.15mm T with G.I coated IS2629 std)

c) Fan Drive/motor suspension structural beam (The fan bearing block is resting on these beams of ISMB 250.These beams needs to be replaced with new beam and Fabrication of the same will have to be done with high precision particularly with respect to the level and center position of the plenum chamber as the fan is mounted on this beam. Vendor to note the same)
The motor stand and bearing block standard belt tension screw assembly to be checked in the site for exact dimensions

d) Fan screen guard shall be fabricated with mesh size/ spec IS 412 (SWM 20,LWM 60,W-3.25 mm,T-2.24mm).(The Fan pulley is protected by a belt guard with approximate size of 1600mmx1350mm.This belt guard needs to be replaced by new belt guard made of angles and mesh as per site measurement.)

e) A Manway of approximate size 750mmx750mm to be fabricated in the bottom mesh with a suitable hinge arrangement for entry and exit of maintenance personnel inside the plenum chamber.

f) All required fasteners / bolts/ studs nuts,washers etc., required for installation /fixing at site.

g) All /Any other structures/ parts shall be as per drawing and agreed scope.

2.2 Procurement of raw materials /bought out items

Contractor shall procure all materials / bought out items for above mentioned/required structural members / parts as marked in the tender drgs & any other structures parts which are to be replaced.as per specification.

In addition to the materials required for complete fabrication, additional 25% of total materials used for plenum chamber assembly fabrication as listed below shall be supplied to site along with final supply for any site modifications during installation.

a) **IS 2062 std, plate 3.15mm T with G.I coated IS2629 std .**
b) **Mesh size/ spec IS 412 (SWM 20,LWM 60,W-3.25 mm,T-2.24mm)**
c) **All required fasteners / bolts/ studs nuts,washers etc.**
   Below listed items to be supplied as per mentioned quantity. Pre-Fabrication required if any, shall be done as per site measurement and EIC instruction.
d) **GI pipe – 1-1/4 class B as per IS1239 for handrail replacement at site : 150 meter**
e) **ISMC 100X50X5mm For : 1000 Kg (1 ton)**

Required test certificates of the raw materials /bought out items shall be submitted to MRPL.

2.3 Fabrication

Carrying out fabrication works of the above mentioned structural members / parts as marked in the tender drgs &other structures parts as per agreed scope (Plenum Chamber with truss assembly, Bottom cover plate with supporting structural members, Fan Drive/motor suspension structural beam Fan wire mesh guard, etc.)

Fabrication shall be as per the tender & drawing dimensions, existing site conditions and instruction of Engineer In Charge (EIC).

Entire Fabrication, welding & painting of the items as per scope shall be carried out at Vendors works site / premises.

Welding of carbon steel plates by qualified welder as per the MRPL approved procedure and relevant codes and standards. The welders used for the welding will have to be approved by MRPL Inspection.

Necessary drilling on the Frame (ISMB, ISMC, Angles) to be carried out to fix the Motor/Fan Assembly, Mesh frame at the bottom of the circular duct and at any other location as per site conditions /measurements.

2.4 Painting

Complete fabricated structures / parts (except for the GI coated parts/surfaces) as per tender scope, to be painted by the contractor with the following specification (MRPL specification attached separately)

Surface Preparation:

Hand tool cleaning/Manual Cleaning to SSPC-SP-2 should be done.

Painting :

Primer , one coat of epoxy mastic, 100 microns dft

Intermediate, one coat of epoxy high build, polyamide cured, 100 microns dft

Finish, one coat of acrylic polyurethane aliphatic isocyanate cured, 40 microns dft .

Painting scope includes both supply & application

2.5 Delivery /shifting of fabricated items to MRPL site

Contractor shall complete the fabrication& Painting at their worksite and obtain clearance from MRPL / TPI before shifting to MRPL site.

Vendor shall make the necessary arrangements for shifting of the fabricated materials items as per scope from their works location to delivery inside MRPL Mangalore site (location as per EIC instructions).This scope includes to and fro transportation, necessary Government pass / permits required to carry these materials shifting . Gate pass for MRPL entry shall be arranged by MRPL .Prior information regarding the same shall be intimated by vendor well in advance.

2.6 Site fabrication for Correction / rectification works

If the vendor supplied items dimensions are found to be not matching the tender drawings / approved drgs, Vendor / contractor shall carryout necessary correction / rectification of supplied Fin fan section/ parts and also in the existing finfan structures /parts, so that the installation is completed in required manner.

Any other Rectification works required as per vibration / condition monitoring report (for reasons assessed due to Dimensional variations /non-confirmation of the supplied items with respect to the As-built drawings) shall also to be carried out by vendor (post installation & trail run).

2.7 During /after completion vendor's work scope at MRPL site, vendor shall Carryout housekeeping of the work place and return all scrap material to MRPL scrap yard as per the instruction of EIC.MRPL will provide Truck and crane with necessary fuel and driver on free of cost basis for shifting the removed materials to MRPL scrap yard or to a location as indicated by EIC

2.8 The contractor may visit the site before quoting for the job.

2.9 Contractor shall complete the fabrication at their worksite and arrange resources well in advance for carrying out the Supervision & other works as per scope during Shutdown of HCU2 unit or as instructed by EIC.

2.10 The Contractor shall submit a tentative schedule before starting the job and carry out continuous progress monitoring till completion of the job.

3.0 SCOPE OF SUPPLY:

3.1 Vendor's Scope of Supply:

a) Procurement of all materials required as per tender scope.
b) Fabrication (at vendor's work site)
c) Painting (Supply & application) (at vendor's work site)
d) For Site fabrication for Correction / rectification works.

Sufficient Welding Machines with adequate power &welding cables, Grinding Machines, sufficient Fire blanket, Cutting Sets, Lifting tools(other than crane), Tools and Tackles , etc and all other required for execution & completion of this Site fabrication for Correction / rectification works .These items brought inside MRPL by the contractor can be taken back by the contractor after completion of the work order.

Consumables like welding electrodes (as per list of approved vendor list) , filler wires, oxygen ,acetylene or cutting gas, grinding wheels(as per list of approved grinding wheels stated in tender),etc required during the above activities.

All personnel protective and safety equipments like gloves, double lined full body harness safety belts, safety shoes, goggles, helmets, ear plugs, dust masks and welding related personal protective equipments etc required for the safe execution of the job.The contractor shall be solely responsible for ensuring that personal protective safety items are used throughout the day while on duty and also for the safety & health of their employees/ workmen .These PPE brought inside MRPL by the contractor can be can be taken back by the contractor after completion of the work order.

3.2 MRPL'S Scope of Supply:

a) Truck , hydra, Crane with necessary fuel and drivers on free of cost basis, required for unloading supplied items, shifting the supplied item from MRPL Stores to fabrication site or any area allotted to contractor to carry out the Site Correction / rectification works of the supplied parts(within MRPL premises).

b) Replacement / Installation of the supplied parts of the Finfans at MRPL site (through MRPL own/ other Contract resources)

c) Water, Compressed Air and Electricity required for carrying out jobs at MRPL site shall be provided free of charge basis(for Site fabrication for Correction / rectification works)

4.0 SPECIAL CONDITIONS OF CONTRACT:

a) MRPL General Purchase Conditions shall be a part of this contract. The Vendor/contractor is advised to go through the same and comply with the conditions.

b) The Vendor/contractor has to ensure deployment of necessary manpower with relevant experience /skilled manpower, to carry out Job of this nature

c) Contractor shall comply with following terms &conditions for any MRPL site works (for Site fabrication for Correction / rectification works)

- The Contractor shall strictly follow safety regulations, permit systems & housekeeping in the area of work at MRPL site.
- The Contractor has to ensure deployment of necessary manpower with relevant experience /skilled manpower, to carry out Job of this nature at MRPL site. Key Resource Persons viz., Site in-charges, Supervisors, safety persons qualifications & experience shall be as per Annexure A to this tender scope of work. Confirmation of the same shall be submitted by the bidder along with the Bid . Necessary Documentary proof for the same shall be submitted prior to mobilization at Site .
- Non – mobilization/ Deployment of such Key Resources Persons at MRPL site during execution as prescribed above shall attract penalties as described in Penalty clause in this tender document Deployment of Welders& Fabrication crew:.
- Contractor shall deploy sufficient qualified welders along with adequate nos of skilled grinders and fabricators at MRPL site for executing tender scope of work meeting the time schedules and work quality.
- Deficiency in mobilization as prescribed above shall attract penalties as described in Penalty clause as follows

  Penalty for Non-mobilization/ Deployment of Key Resource Persons per day per person after the contractual mobilization period/ schedule (for MRPL Site works) shall attract following penalties

  Rs. 5000/- for Site-In-Charge

  Rs 3000/- for Supervisor/ Discipline Engineer

  -For Safety Officer / Supervisor - as per MRPL Contract Safety policy.

  Penalties for safety violations shall be levied as per MRPL Contract Safety policy

  Safety Violation Penalty: The Contractor shall strictly follow safety regulations inside the Refinery. In case the Contractor is found to violate any safety norms like PPE violation, unsafe act/ condition or any near-miss is reported or Reportable Lost Time Injury is reported, the penalty as per "Contract Workers Safety policy" will be levied. The decision of EIC will be final in this regard.
- PPEs: All Personnel deployed by Contractor for this job should be provided with all PPE as per the permits/safety requirements.
- Deployed contractor's personnel shall maintain good discipline, safety standards, commensurate with industrial standard /job requirement. In the opinion of the EIC, any of the contractor personnel found unsuitable for services shall be discontinued and the contractor shall provide suitable alternate personnel immediately.
- The job should be carried out with a valid work permit only and the contractor to ensure that there is no violation of the conditions as mentioned in the permit.

Contractor shall observe all safety precautions and obtain necessary Work permits before carrying out any job inside the Refinery Complex. The contractor shall strictly adhere to all conditions and safety precautions as mentioned in the work permits. MRPL reserves the right to cancel any hot work permits issued without assigning any reason. Contractor shall educate his team to stop immediately the hot jobs in the event of fire.

- Contractor's deployed workmen/personal to be paid as per the latest HR circular on payment to secondary workforce and contractor to strictly comply with this condition.
- Contractor shall be responsible for the safety and health of all his employees. All liabilities under IE Rules 1956/ Labour laws, Insurance on account of this contract for personnel / labours shall be done by the contractor.
- It shall be ensured that the Manpower deployed by the Contractor for the subject works shall be in two separate crews to cover a 24-hours day work, with each crew strictly not exceeding 12 hours. Contractor to ensure one day weekly off to the deployed personnel after 6 days of continuous work.
- For all identified critical jobs (which will be communicated separately by MRPL) contractor should deploy dedicated qualified site safety supervisor.
- All critical activities shall be carried out under proper supervision .In case the supervisor is leaving the site during site work execution, it shall be ensured that the job is brought to a safe condition prior to leaving the work place and no activity shall be carried out in absence of supervisor.
- Toolbox talks (TBT) to include job specific topics which shall also address potential human error possibilities and other related hazards.
- All JSA (if made) guidelines /instructions and Work related SOPs to be followed without deviation.
- The Contractor shall provide the contact address and mobile number of the Supervisor/Site In charge for any Emergency. Supervisor &Site In charge shall always possess a working Intrinsic-safe mobile phone for this purpose during execution of the contract work scope.
- The contractor shall be ensured that two crews cover a 24-hours day work with each crew not exceeding 12 hours. Contractor to ensure one day weekly off to the deployed personnel after 6 days of continuous work.
- The Contractor shall return all special safety equipment and other items taken from MRPL on completion of work duly certified by the Engineer-in-charge. For any items lost or damaged by the Contractor, the replacement cost shall be recovered from the Contractor's bills.
- Entry of Man/material/equipment shall be permitted within the Refinery area with a valid pass and no material/equipment shall be permitted to be taken out of the Refinery area, unless authorized by the concerned authorities of the Refinery with valid gate pass. The Contractor shall be held fully responsible for any or all delays/losses/damages that may result consequent on any lapse(s) that may occur on the part of his employees in this regard. All Materials, equipments, machineries, tools, etc., shall be brought inside by the contractor vide returnable gate passes & can be taken back by the contractor after the completion of the jobs under the contract.
- MRPL will arrange to issue entry passes/Bio-Metric punch cards to the Contractor for every workmen deployed by the Contractor and would made valid for the estimated period of work. Such cards shall be returned after the period is over. They are also required to punch in and out every time at the Gate while they enter or leave the refinery.

- The contractor has to obtain a police verification clearance certificate for the resources deployed by the contractor.
- Contractor should get his all workmen / supervisor medically examined at his cost and the medical certificate to be submitted to relevant Department in MRPL . Also necessary medical tests shall also be carried out by Contractor for their Workmen/ employees to be Deployed by Contractor for height works &drivers during the contract execution period, with his own cost, as specified in the attached medical fitness/ checkup format.
- The Contractor shall ensure that all industrial gases such as Oxygen, Acetylene, Bharat cutting gas, Argon, Nitrogen, welding electrodes/filler wires are pre-approved by MRPL before they are used for the job.
- Before commencement of the works at site, Contractor shall prepare and submit Standard work procedure (SWP) / Standard Operating procedure (SOP) of their works scope as per tender ,incorporating all step by step work procure with precautions & safety requirements to MRPL /EIC for review & approval .The same shall be followed and Contractor shall ensure that their deployed crew is following SWP/SOP. Daily /shift-wise tools box meeting/talks shall include such instructions.
- The welders required to perform the job will have to be qualified by MRPL Engineer-in-Charge. The cost of Qualification Tests will have to be borne by the Contractor. Required test piece s will be issued to contractor by MRPL at free of cost
- All temporary facilities required will have to be arranged by the contractor and such materials brought inside on returnable basis can be taken back by the contractor after the completion of the contract with relevant gate passes.
- Electrodes and filler rods to be properly stored and baked as per requirement. The electrodes/filler wires shall be with batch test certificates and electrodes if found not fit the Contractor shall replace them with good quality electrodes.
- All welding work equipment for welding and other auxiliary functions and the welding personnel should meet the requirement laid down in the latest editions of the standards and procedures.
- Contractor to quote as per the Schedule of Rates. The quantities mentioned in SOR are indicative only and the payment will be made as per the quantities executed and certified by Engineer In Charge. The contractors are advised to go thro' all the attachments carefully before quoting for the job. For any clarification/doubt EIC may be contacted.
- Contractor shall visit the site before quoting for the job and clarify doubts, if any, with the EIC.
- Boarding, lodging for contractor personnel required during the execution of the job is in the scope of the contractor only.
- The necessary transportation for the contractor men, tools and equipment during the course of the job to be completely arranged by the contractor only.
- Food, tea, snacks for the contractor personnel to be arranged by the contractor at his own cost. A canteen for contract employees is available for Tea, Snacks, and Lunch & Dinner for their employees during the job inside the refinery.

d) SAFETY INSTRUCTIONS TO BE STRICTLY FOLLOWED BY CONTRACTOR(For MRPL SITE WORKS)

- All welding machines Power connection should be connected to the welding receptacle through welding plug tops only.

- Supply power cable to welding machines, welding current regulator, portable electrode oven, grinding machines, power distribution board etc. should not have any cable joints. (Single piece Cable).
- Welding Cables & Gas hoses should be inspected for cuts, Leaks, Brakes & Insulation damages. The Fittings & Valves of Gas & Oxygen Pressure Cylinders should be inspected for leaks.
- Pressure Cylinders should be kept at a safe distance from welding or cutting operations.
- All power cable ends should have industrial plug on one side and other end directly into the machine. (No naked end pinning into will be permitted)
- Earthing of welding cable should be rigidly connected to the material being welded & securely attached at a location immediately adjacent to the welding.
- Welding cable ends should be lugged & bolted on the machine side, Holder & earthling side. No joints will be permitted on the welding cable. Only aluminum/ copper cable should be used for welding holder & earthling during welding.
- Grinding machines should be connected through a three core single cable with industrial plug top one side and direct to the machine on the other end. (No cable joint will be permitted).
- All grinding machines used should have wheel guards.
- No jobs should be started without the valid work permit and to be stopped on expiry/withdrawal of the permit. When the welder stops working, the welding machine should be shutdown & the valves on the cylinders should be closed and the pressure from the regulators should be released.
- All stand by fire fighting equipment as mentioned in the hot work permit to be ensured at the place of work.
- All the workmen & Supervisor should have personal protective equipments like Helmet, Safety Shoes, gloves, Welding helmet & gloves for the welder, Goggles for Grinder & Gas Cutter. Safety belts with double harness should be used while workingatheights.The contractor is solely responsible to ensure that his workmen use the proper PPE as per work permit requirements
- The Contractor shall get the Welding machines certified by any of the MRPL approved certification agency. The Cost towards the certification will have to be borne by the contractor and no additional charges will be paid by MRPL.
- When the welder stops working, the welding machine should be shutdown & the valves on the gas cylinders should be closed and the pressure from the regulators should be released

5.0 DELIVERY SCHEDULE /COMPLETION PERIOD

For Supply: Contractor shall complete the entire supply scope of the contract including fabrication & delivery at MRPL site within **03 (three) months** from date of placement of PO.

For Site fabrication for Correction / rectification works: Contractor shall arrange Manpower & other resources well in advance for carrying out MRPL site works (For Site fabrication for Correction / rectification works) as per scope during Shutdown of HCU2 unit or as instructed by EIC. The Contractor will be given a notice period of **FIVE days** for mobilization of resources and the contractor shall commence the work accordingly as per the instruction of Engineer-in-charge.
The replacement job (by MRPL) is planned to carryout during HCU2 unit shutdown, tentatively in April-May 2025. Exact shutdown period will be intimated well in advance.
Contractor shall complete all the Site works at MRPL as per tender scope within **10(Ten) days** from date of clearance for site work, as communicated by MRPL. (Installation in MRPL scope)

6.0 SCHEDULE OF RATES:

JOB DESCRIPTION: Supply of Plenum Chamber & other parts of Fractionator section Fin fans in HCU2 for replacement of damaged sections.

SL No	DESCRIPTION	UOM	QTY	Quoted Price (per unit)
1	Supply of replacement section /parts of EC42101 scope includes Procurement, fabrication , painting & delivery of Fin fan cooler Plenum Chamber with truss assembly, Bottom cover plate with supporting structural members, Fan Drive/motor suspension structural beam, Fan wire mesh guard and all required fasteners etc., as per specifications and as per the detailed tender scope of supply.	**LUMPSUM PER FAN**	**6**	
	Total Amount (INR)			

NAME :
Company Name : ____________________
Address : ____________________________
____
Phone : ________________
Fax : __________________
Email : ______________

1. The Vendor /Contractor will be paid based on rates quoted ,duly certified by the Engineer-in-charge, as per PO Payment terms

7.0 LIST OF APPROVED ELECTRODES

1. AdvaniOerlikon (Bhandup/Chennai/Ahmednagar/Mumbai)
2. EsabIndia (ambattur/kalwa/khardhah)
3. D&H
4. Modiarc (Modinagar)
5. Honavar(Badlapur/Thane)
6. Kobe steel (Timba,Gujarat)
7. Mailammetallogen(Pondicherry)

8.0 LIST OF APPROVED GRINDING/CUTTING WHEELS

1.Carborandom universal
2.Grindwel Norton
3.Royal Arc Abrasives
Other items shall be of reputed brand/good quality and this shall be accepted by EIC.

9.0 LIST OF DRAWINGS ATTACHED

1) Drawing No. 189811000 rev 2 for ec42101.
2) EC42101 three dimensional sketch of the fin fan cooler bottom duct.
3) BOM details of EC42101.

10 PAYMENT TERMS

PAYMENT TERMS			
	SOR item no / Description	**UoM**	**Payment terms**
1	SOR item no **S**upply of replacement section /parts of EC42101 scope includes Procurement, fabrication , painting & delivery of Fin fan cooler Plenum Chamber with truss assembly, Bottom cover plate with supporting structural members, Fan Drive/motor suspension structural beam, Fan wire mesh guard and all required fasteners etc., as per specifications and as per the detailed tender scope of supply**.**	Lum psum	**80%** of Lumpsum amount upon completion of all of the following as per tender scope • Procurement of all bought out items and submission of its receipt , required Test certificates. • Fabrication and Painting of the Fin fan items as per scope /drg • Shifting of the supply scope items to MRPL site and acceptance of material by MRPL **20%** of Lumpsum amount upon completion of all of the following • Necessary correction / rectification works of supplied Fin fan section/ parts ,if required as per Observations during installation at site . • Trail run /commissioning of the Fin fan Equipment • Clearing of scraps& other Contractor's own material from the MRPL site and issuance of completion certificate by MRPL.

KEY RESOURCE PERSONS QUALIFICATION & EXPERIENCE (POST QUALIFICATION) (FOR WORK EXECUTION AT MRPL SITE)

CATEGORY	**QUALIFICATION & EXPERIENCE (POST QUALIFICATION) REQUIRED**	
Site-In-Charge	Degree or Diploma in Engineering with minimum relevant experience as given below	
	Degree holders	5yrs
	Diploma holders	8yrs
Supervisor/ Discipline Engineer	Degree or Diploma in Engineering with minimum relevant experience as given below	
	Degree holders	2 yrs
	Diploma holders	4 yrs
Safety Officer / Supervisor	As per MRPL Contract workers safety policy attached as part of tender document	

MEDICAL EXAMINATION FOR WORKERS WORKING AT HEIGHT

(FOR WORK EXECUTION AT MRPL SITE)

Name :Blood Group:	
DOB :Age:Sex: Male/ Female	
Designation:	Married/ Unmarried
Nature of work: Work Duration: Address:	

GENERAL EXAMINATION:

Pulse Rate:_____/min BP:___/___mm of Hg SPO2: ____% Resp. rate:____/min

Weight :______kgs Height: ______cms Chest: Inspiration: _______cms

Expiration: _____cms Abdominal girth: ________cms

Pallor: Icterus: Cyanosis: Clubbing:

Lymphadenopathy: Edema:

MEDICAL HISTORY:

H/O Fear of Heights:

H/O Vertigo/Giddiness:

H/O Epilepsy:

H/O DM/HTN/CHD:

H/O Alcohol/Drug abuse:

MEDICAL EXAMINATION FOR WORKERS WORKING AT HEIGHT

H/O Accidents: H/O any other major illness/ hospitalization: H/O Vision complaints: H/O Hernia/ any Surgical interventions: H/O Hearing complaints: Other Complaints:

PHYSICAL EXAMINATION:

1.Eyes: Near vision : Distance vision: Color vision :	Right	Left
2. Ears: Hearing test by tuning fork: Any middle ear disease :		
3. Nose:		
4. Throat:		
5. Skin:		

MEDICAL EXAMINATION FOR WORKERS WORKING AT HEIGHT

6. CVS:	Heart sounds:	Murmurs:
7. Respiratory system:	Breath sounds:	Other sounds:
8. Gastro Intestinal System:		
9. CNS:	Reflexes:Nystagmus: Romberg`s sign:	 Finger to Nose test:
10. Spine:		
11. **Examination for physical deformities** (to assess the ability to climb): Limbs:Rt(U) Lt(U) Rt(L) Lt(L) **Remarks:**		

LABORATORY INVESTIGATIONS:

HB: _____gm% TC:_____ Cumm ESR: ________mm/hr Platelets: ______($x10^9$/L) DC: N:______% L:______% E:______% M:______% FBS/RBS: _________mg/dl Blood Group:________ S. Creatinine:________mg/dl
URINE ROUTINE: Microbiology: Albumin: Sugar:

MEDICAL EXAMINATION FOR WORKERS WORKING AT HEIGHT

Chest X-RAY (P A view):
ECG:
Audiometry (Yearly once):
Breath alcohol test (if required)

OPINION OF FITTNESS: FIT / UNFIT / TEMPORARILY UNFIT:
REMARKS:

If Medical examination not done at MRPL Hospital:

Medical examination done at______________________________(Name & Address of Hospital)

(Authorized Signatory of Hospital)

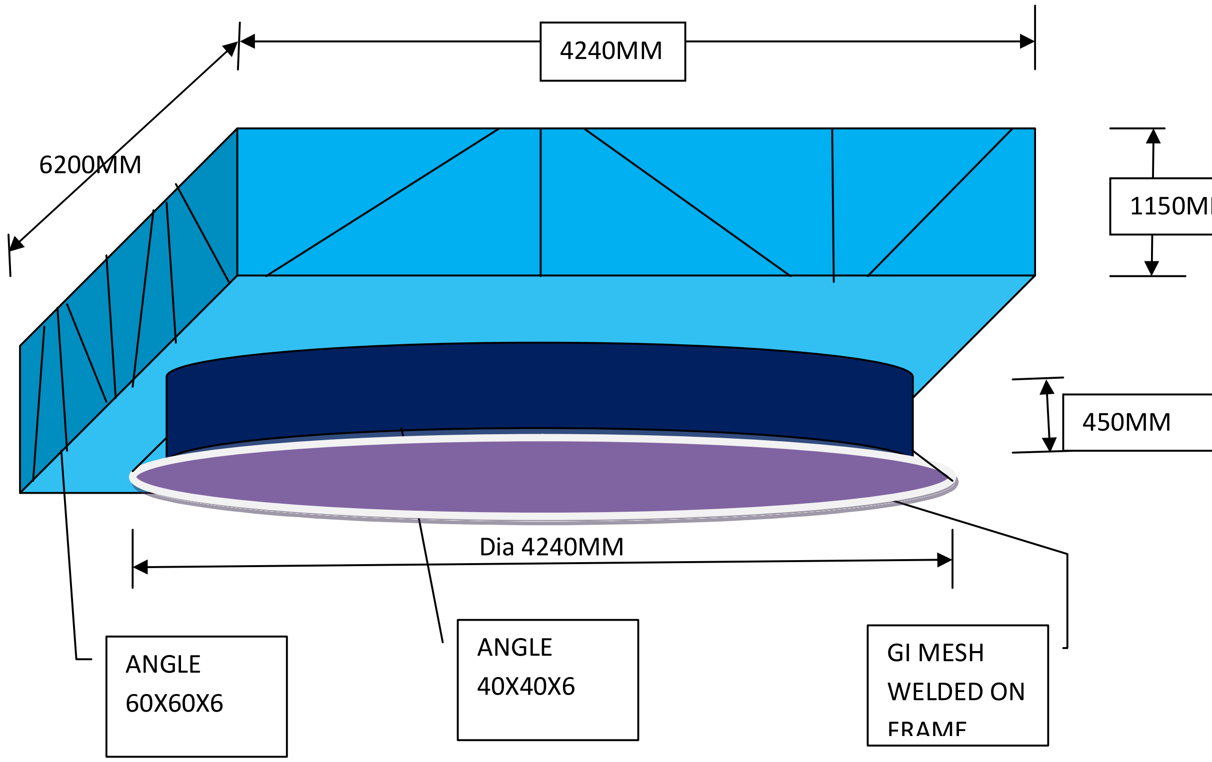
Place:

Date:

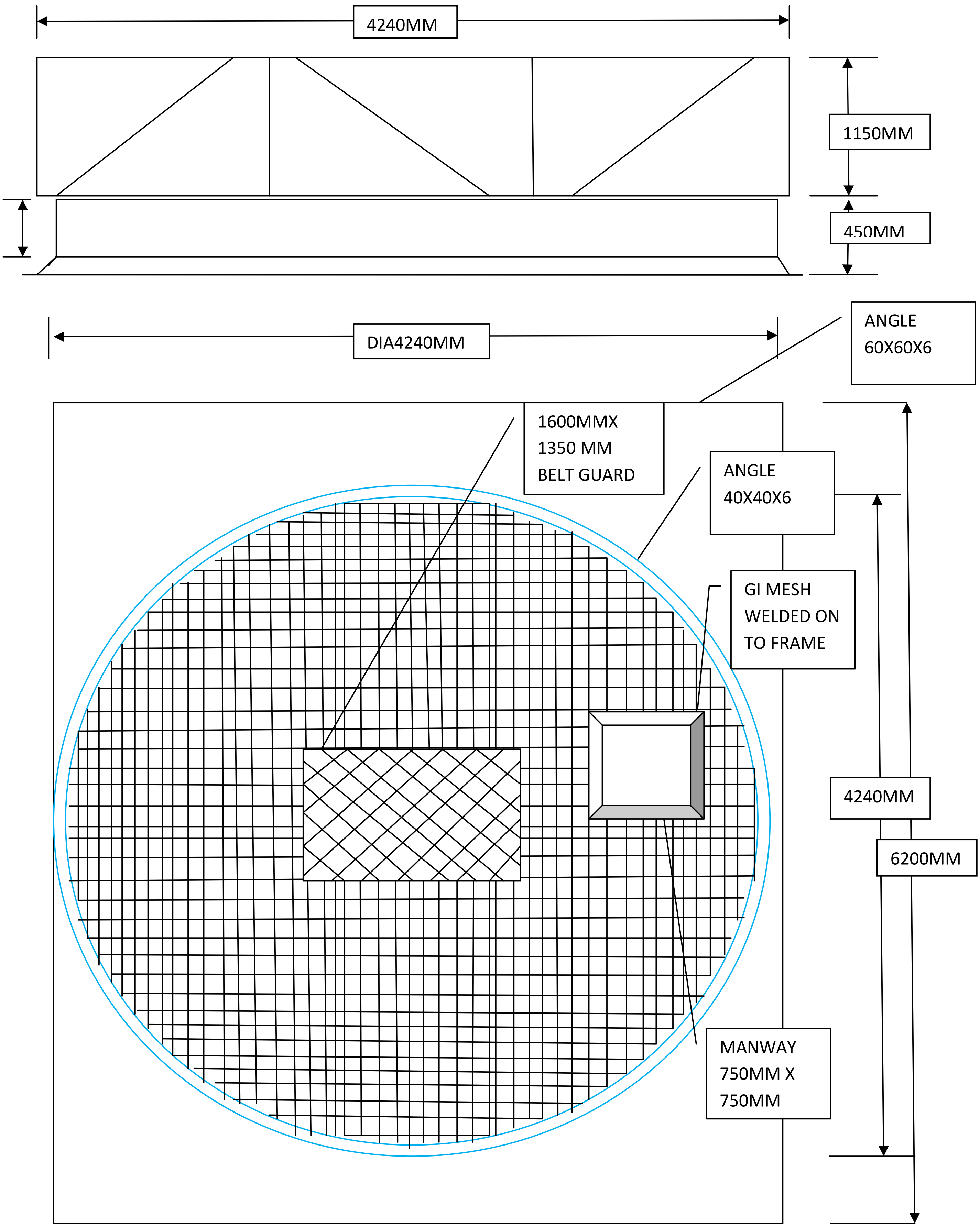
Registered Medical Practioner
Sign & Seal

EC 42101 THREE DIMENSIONAL SKETCH OF THE FIN FAN COOLER BOTTOM DUCT FOR ESTIMATION PURPOSE ONLY

(Site dimensions to be verified by contractor before fabrication)



DRAWING SHOWING THE DIMENSIONS FOR ESTIMATION PURPOSE ONLY AND NOT FOR FINAL FABRICATION



BOM details of EC42101

SL NO	ITEM NO.	DESCRIPTION	SIZE	IDENTI.	DRG. NO	QTY	TAG NO.
1	EC-42101	COLUMN	ISMB 300	TYPE A-24.1A	18981122529	1 NOS	E C 42101(Q&R)
2	EC-42101	COLUMN	ISMB 300	TYPE A-24.1B	18981122529	1 NOS	E C 42101(Q&R)
1	EC-42105	COLUMN	ISMB 300	TYPE A-13 A	189811228	1 NOS	E C 42105(A&B)
2	EC-42105	COLUMN	ISMB 300	TYPE A-13 B	189811228	1 NOS	E C 42105(A&B)
3	EC-42105	COLUMN	ISMB 300	TYPE A-13.1A	189811228	1 NOS	E C 42105(A&B)
4	EC-42105	COLUMN	ISMB 300	TYPE A-13.1 B	189811228	1 NOS	E C 42105(A&B)

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SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
C	DRIVE ASSY. AND MOTOR SUSPENSION						
33	MAIN BEAM ASSY.	ISMB 250 (1550 X 8650)	---	1898 11 320	1 NO	9 NOS	EC42101(A T
34	MOTOR STAND	ISA	---	1898 11 320	2 NOS	18 NOS	EC42101(A T
35	BEARING BLOCK STAND	420 X 1550	---	1898 11 320	2 NOS	18 NOS	EC42101(A T
36	TENSIONING SCREW ASSY	---	--- ---	1898 11 320 1898 11 320	2 NOS	18 NOS	EC42101(A T EC42101(A T
D	BEARING BLOCK	12 FT DIA FAN	---	1898 11 327	1 NO	18 NOS	EC42101(A T
E	FAN GUARD						
37		S CIRCLE	---	1898 21 377	2 NOS	36 NOS	EC42101(A T
38		SQUARE	---	1898 21 377	---	54 NOS	EC42101(A T
F	FAN RING & CONE ASSY						
39		DIA 12 FT		1898 11 370	1 NO	18 NOS	EC42101(A T

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
G	STRUCTURE ASSY.						
40	COLUMN	ISMB 300	TYPE A 9.1	1898 11 221	1 NO	---	EC 42102(A)
41	COLUMN	ISMB 300	TYPE B 9.1	1898 11 221	1 NO	---	EC 42102(A)
42	COLUMN	ISMB 300	TYPE B10.1	1898 11 221	1 NO	---	EC 42102(B)
43	COLUMN	ISMB 300	TYPE A10.1	1898 11 222	1 NO	---	EC 42102(B)
44	TRUSS	1300 X 8100	TR 06	1898 11 231	2 NOS	---	EC 42102
45	TRUSS	1300 X 5572	TR 01A	1898 11 234	1 NO	---	EC 42102
46	TRUSS	1300 X 5762	TR 01	1898 11 230	2 NOS	---	

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
47	BRACING	ISA 80X80X8	TYPE 1	1898 11 227	4 NOS	- -	EC 42102
48	BRACING	ISA 80X80X8	TYPE 6	1898 11 227	2 NOS	---	EC 42102
49	BRACKET	ISMB 250	TYPE 1	1898 11 236	1 NO	---	EC 42102
50	BRACKET	ISMB 250	TYPE 2	1898 11 236	1 NO	---	EC 42102
51	BRACKET	ISMB 250	TYPE 3	1898 11 236	2 NOS	---	EC 42102
52	BRACK[illegible]	ISMB 250	TYPE 5	1898 11 [illegible]	2 NOS	---	EC 42102&103
53	TRUSS BOTTOM COVER	---	---	1898 2 1232	2 NOS	---	EC 42102
54	PLATFORM	900 X 11650	---	1898 11 236	1 NO	---	EC 42102
55	PLATFORM	900 X 6930	---	1898 11 236	1 NO	---	EC 42102
56	PLATFORM	1800 X 10750	---	1898 11 236	1 NO	---	EC 42102 & 10[illegible]
57	PLATFORM	900 X 8250	---	1898 11 236	1 NO	---	EC 42102 & 10[illegible]
H	**DRIVE ASSY. AND MOTOR SUSPENSIO N**						
58	MAIN BEAM ASSY.	ISMB 250 (1550 X 8850)	---	1898 11 321	1 NO	---	EC 42102
59	MOTOR STAND	ISA	---	1898 11 321	1 NO	2 NOS	EC 42102
60	BEARING BLOCK STAND	420 X 1550	---	1898 11 321	1 NO	--- 2 NOS	EC 42102
61	TENSIONING SCREW ASSY	---	---	1898 11 321	1 NO	2 NOS	EC 42102
I	**BEARING BLOCK**	12 FT DIA FAN	---	1898 11 325	1 NO	2 NOS	EC 42102
J	**FAN GUARD**						
62		S CIRCLE	---	1898 21 372	2 NOS	4 NOS	EC 42102
63		SQUARE	---	1898 21 373	6 NOS	---	EC 42102
K	**FAN RING & CONE ASSY**						
64		DIA 12 FT		1898 11 370	1 NO	---	EC 42102

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
L	STRUCTURE ASSY.						
65	COLUMN	ISMB 300	TYP B 10.2	1898 11 222	1 NO	---	EC 42103(A)
66	COLUMN	ISMB 300	TYP A 10.2	1898 11 222	1 NO	---	EC 42103(A)
67	COLUMN	ISMB 300	TYP B 11.1	1898 11 223	1 NO	---	EC 42103(B
68	COLUMN	ISMB 300	TYP A11.1	1898 11 223	1 NO	---	EC 42103(B
69	COLUMN	ISMB 300	TYP B12.1	1898 11 228	1 NO	---	EC 42103(D)
70	COLUMN	ISMB 300	TYP A12.1	1898 11 228	1 NO	---	EC 42103(D)
71	TRUSS	1300 X 8100	TR 08	1898 11 231	3 NOS	---	EC 42103
72	TRUSS	1300 X 5572	TR 02	1898 11 230	4 NOS	---	EC 421[illegible]
73	BRACING	ISA 8[illegible]X8	TYPE 2	1898 11 227	8 NOS	---	EC 42103
74	BRACING	ISA 80X80X8	TYPE 6	1898 11 227	4 NOS	---	EC 42103
75	BRACKET	ISMB 250	TYPE 3	1898 11 238	2 NOS	---	EC 42103
76	BRACKET	ISMB 250	TYPE 5	1898 11 238	2 NOS	---	EC 42103 & 1[illegible]
77	BRACKET INCLUDED IN EC42102		TYPE 5	1898 11 238	---	---	---
78	TRUSS BOTTOM COVER	---	---	1898 2 1232	4 NOS	---	EC 42103
79	PLATFORM	1800 X 10750	---	1898 11 236	1 NO	---	EC 42103 & 10
80	PLATFORM	1400 X 2200	---	1899 11 236	1 NO	---	EC 42103 & 10
81	PLATFORM INCLUDED IN EC 42102		---	---	---	---	---
M	DRIVE ASSY & MOTOR SUSPENSION						
82	MAIN BEAM ASSY.	ISMB 250 (1550 X 8850)	---	1890 11 322	1 NO	2 NOS	EC 42103
84	MOTOR STAND	ISA	---	1898 11 322	1 NO	2 NOS	EC 42103
85	BEARING BLOCK STAND	420 X 1550	---	1898 11 322	1 NO	2 NOS	EC 42103
86	TENSIONING SCREW ASSY	---	---	1898 11 322	1 NO	2 NOS	EC 42103

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
Q	STRUCTURE ASSY.						
90	COLUMN	ISMB 300	TYP B 24.2	1898 11 226	1 NO	---	EC 42104
91	COLUMN	ISMB 300	TYP A 24.2	1898 11 226	1 NO	---	EC 42104
92	COLUMN	ISMB 300	TYP B 25	1898 11 227	1 NO	---	EC 42104
93	COLUMN	ISMB 300	TYP A 25	1898 11 227	1 NO	---	EC 42104
94	TRUSS	1300 X 8100	TR 05	1898 11 231	3 NOS	---	EC 42104
95	TRUSS	1300 X 5572	TR 02	1898 11 230	4 NOS	---	EC 42104
96	BRACING	ISA 80X80X8	TYPE 5	1898 11 227	4 NOS	---	EC 42104
97	BRACING	ISA 80X80X8	TYPE 6	1898 11 227	2 NOS	---	EC 42104
98	BRACKET	ISMB 250	TYPE 2	1898 11 238	2 NOS	---	EC 42104
99	BRACKET	ISMB 250	TYPE 4	1898 11 238	1 NO	---	EC 42104
100	BRACKET INCLUDED IN EC42101		TYPE 6	1898 11 238	---	---	---
101	PLATFORM PLATFORM INCLUDED IN EC 42101		---	---	---	---	---

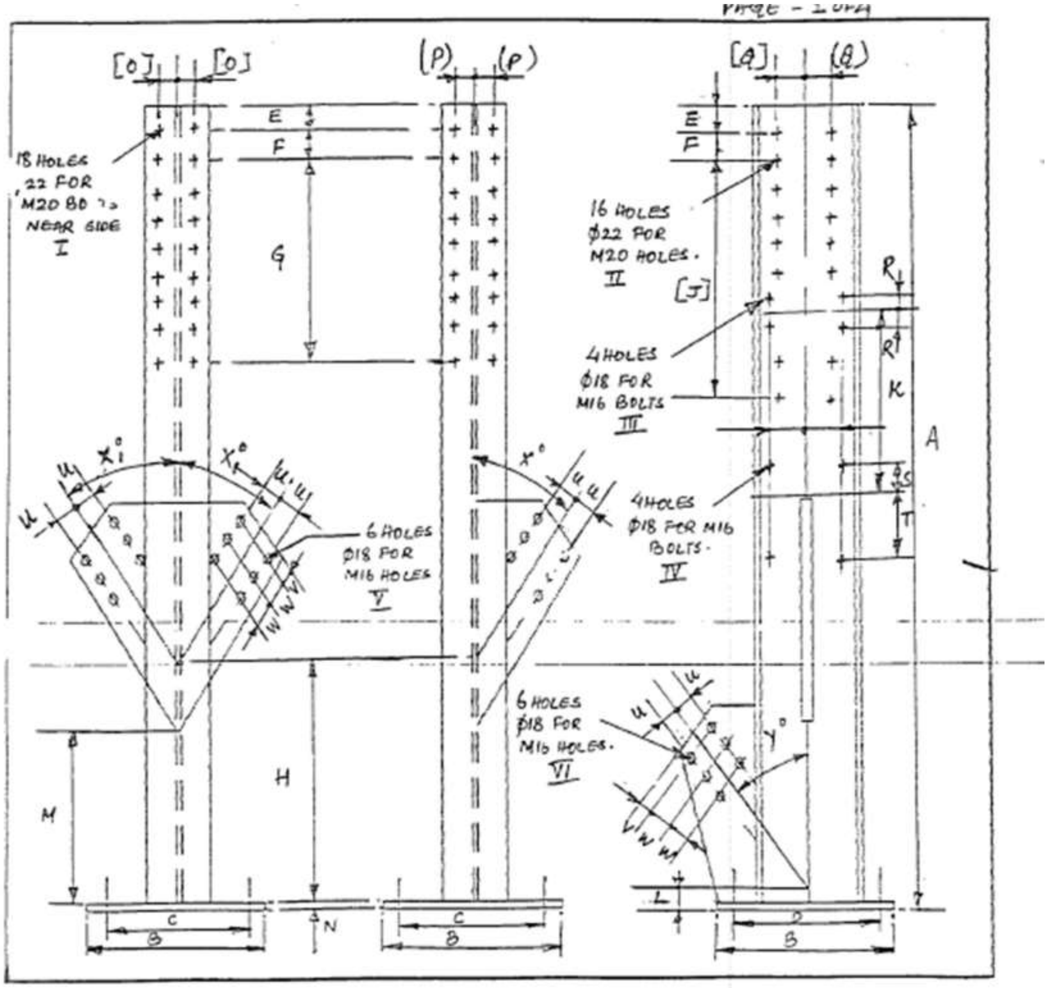
ITEM : M.R.P.L. MANGALORE ... (EC 42104)
ASSY.DRG NO : 189811000

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG
R	DRIVE ASSY & MOTOR SUSPENSION						
102	MAIN BEAM ASSY.	ISMB 250					
103	MOTOR STAND	(1550 X 8650) ISA	—	1898 11 323	1 NO	---	EC 42104
104	BEARING BLOCK STAND	420 X 1550	—	1898 11 323	1 NO	2 NOS	EC 42104
105	TENSIONING SCREW ASSY	—	—	1898 11 323	1 NO	2 NOS	EC 42104
				1898 11 323	1 NO	2 NOS	EC 42104
S	BEARING BLOCK	10 FT DIA FAN	---	1898 11 327	1 NOS	2 NOS	EC 42104
T	FAN GUARD						
106		S CIRCLE	---	[illegible]	4 NOS	---	EC 42104
107		SQUARE	---	1898 21 377	6 NOS	---	EC 42104
U	FAN RING & CONE ASSY						
108		DIA 10 FT	---	1898 11 381	2 NOS	---	EC 42104
[illegible]	[illegible]						

ITEM : RECYCLE TO STORAGE AIR COOLER (EC 42105)
ASSY.DRG NO : 189811001

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
V	STRUCTURE ASSY.						
109	COLUMN	ISMB 300	TYP B 14	1898 11 228	1 NO	---	EC 42105
110	COLUMN	ISMB 300	TYP A 14	1898 11 228	1 NO	---	EC 42105
111	TRUSS	1300 X 8100	TR 06	1898 11 231	3 NOS	---	EC 42105
112	TRUSS	1300 X 5452	TR 02 A	1898 11 234	2 NOS	---	EC 42105
113	TRUSS	1300 X 4112	TR 03 A	1898 11 234	2 NOS	---	EC 42105
114	TRUSS	1300 X 4302	TR 03	1898 11 230	4 NOS	---	EC 42105
115	BRACING	ISA 80X80X8	TYPE 3	1898 11 227	8 NOS	---	EC 42105
116	BRACING	ISA 80X80X8	TYPE 6	1898 11 227	4 NOS	--	EC 42105
117	BRACKET	ISMB 250	TYPE 3	1898 11 238	3 NOS	---	EC 42105
118	BRACKET INCLUDED IN EC42103		TYPE 8	1898 11 238	---	---	---

SL NO	DESCRIPTION	SIZE	IDENTI.	DRG.NO.	QTY/ASSY	TOT. QTY	TAG NO.
	DRIVE ASSY. AND MOTOR SUSPENSION						
119	MAIN BEAM ASSY	ISMB 250 (1550 X 8650)	---	1898 11 323	2 NOS	---	EC 42105
120	MOTOR STAND	ISA	---	1898 11 323	2 NOS	4 NOS	EC 42105
121	BEARING BLOCK STAND	420 X 1550	---	1898 11 323	2 NOS	4 NOS	EC 42105
122	TENSIONING SCREW ASSY	---	---	1898 11 323	2 NOS	4 NOS	EC 42105
X	**BEARING BLOCK**	10 FT DIA FAN	---	1898 11 327	2 NOS.	4 NOS	EC 42105
Y	**FAN GUARD**						
123		S CIRCLE	---	1898 21 377	8 NOS	---	EC 42105
124		SQUARE	---	1898 21 377	12 NOS		EC 42105
Z	**FAN RING & CONE ASSY**						
125		DIA 10 FT	---	1898 11 381	4 NOS	---	EC 42105
[illegible]	**LOUVER DETAIL**						
126	FOR EC 42103	2780 X 8910	---	1898 11 720	4 NOS	---	EC 42103
127	FOR EC 42105	2095 X 8910		1898 11 722	4 NOS	---	EC 42105
[illegible]	[illegible]				[illegible]		[illegible]



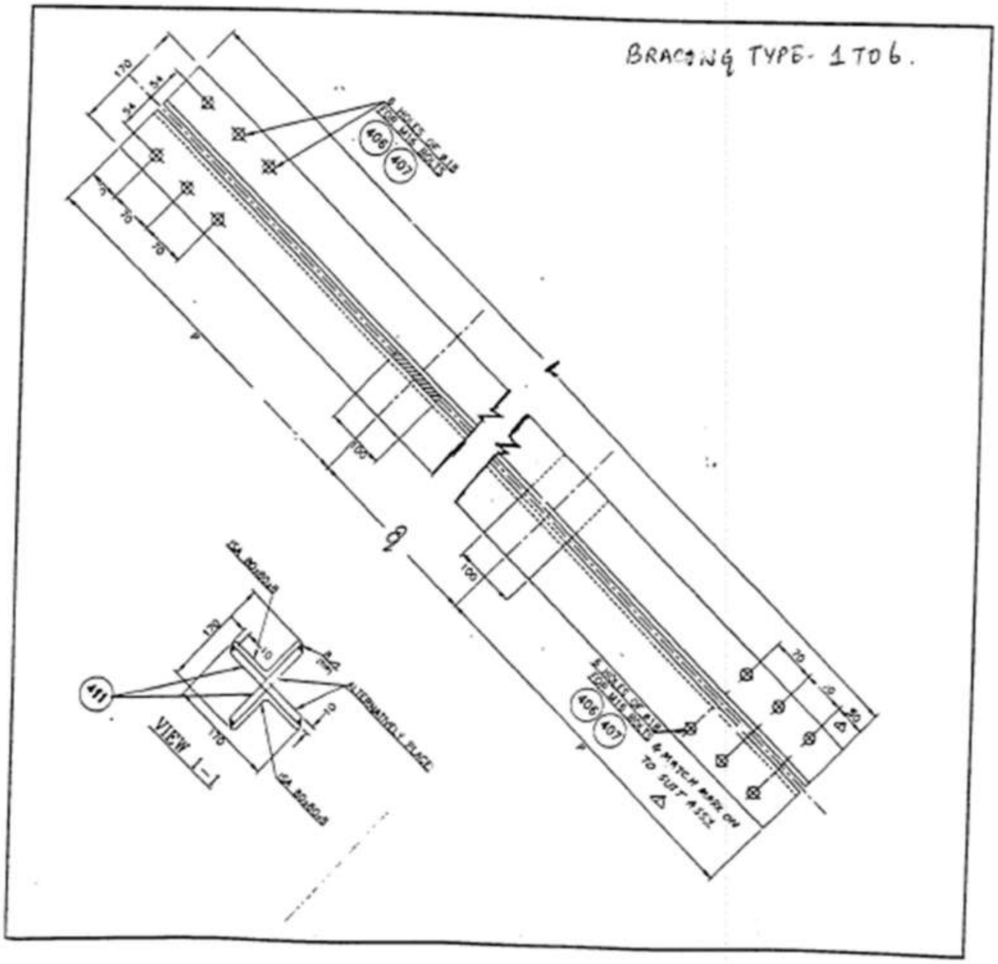
[O]
[O]
(P)
(P)
[Q]
(Q)
E
F
G
H
M
N
C
B
18 HOLES
22 FOR
M20 BOLTS
NEAR SIDE
I
16 HOLES
Ø22 FOR
M20 HOLES.
II
[J]
4 HOLES
Ø18 FOR
M16 BOLTS
III
4 HOLES
Ø18 FOR M16
BOLTS.
IV
6 HOLES
Ø18 FOR
M16 HOLES
V
6 HOLES
Ø18 FOR
M16 HOLES.
VI
A
K
R
S
T
L
D
U
W
V
X°
Y°

DATE OF INSP. :[illegible]

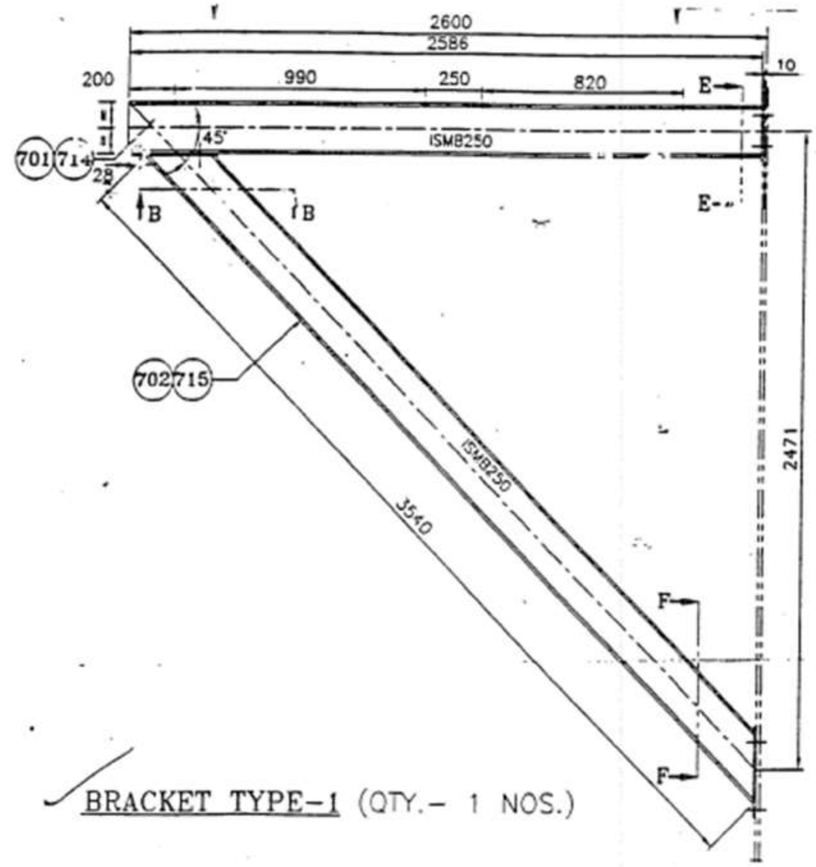
PG-1 OF 1

SL NO.	DESCRIPTION OF ITEMS	REQUIRED DIME.	OBSERVED DIMENSION
1	PLATFORM BETWEEN COL TYP B 9.1 & B 10.1 (LENGTHWISE) - 1 NO	6970mm X 900mm	6970 x 900 (satis factory)
2	PLATFORM BETWEEN COL TYP B 10.1 & B 11.1 (LENGTHWISE) - 1 NO	8250mm X 900mm	8250 x 900 (satisfactory)
3	PLATFORM BETWEEN COL TYP B 11.1 & B 13 (LENGTHWISE) - 1 NO	8250mm X 900mm	8250 x 900 (satisfactory)
4	PLATFORM BETWEEN COL TYP B 13 & B14 (LENGTHWISE) - 1 NO	8620mm X 900mm	8620 x 900 (satis factory)
5	PLATFORM BETWEEN COL TYP B 14 & B 16 (LENGTHWISE) - 1 NO	8764mm X 900mm	8764 x 900 (satis factory)
6	PLATFORM BETWEEN COL TYP B 16 TO B 23 (LENGTHWISE) - 7 NOS	6164mm X 900mm	6164 x 900 (satis factory) - 7 Nos
7	PLATFORM BETWEEN COL TYP B 23 & B 24.1 (LENGTHWISE) - 1 NO	6164mm X 900mm	6164 x 900 (satis factory)
8	PLATFORM BETWEEN COL TYP B 24.1& B 25 (LENGTHWISE) - 1NO	5755mm X 900mm	5755 x 900 (satis factory)
9	PLATFORM BETWEEN COL TYP A 9.1 & B 9.1 (WIDTHWISE)	11050mm X 900mm	11050 x 900 (satis factory)
10	PLATFORM BETWEEN COL TYP A10.1/2 & B 10.1/2 (WIDTHWISE)	11050mm X 1800mm	11050 x 1800 (satisfactory)
11	PLATFORM BETWEEN COL TYPA12.1/13 & B 12.1/13 (WIDTHWISE)	11050mm X 1800mm	11050 x 1800 (satis factory)
12	PLATFORM BETWEEN COL TYP A 14/15 & B 14/15 (WIDTHWISE)	11050mm X 1800mm	11050 x 1800 (satis factory)
13	PLATFORM BETWEEN COL TYP A24.1/2 & B 24.1/2 (WIDTHWISE)	11050mm X 1200mm	11050 x 1200 (satisfactory)
14	PLATFORM BETWEEN COL TYP A 25 & B 25 (WIDTHWISE)	12532/13900mm X 900mm	12532 / 13900 x 900 (satisfactory).

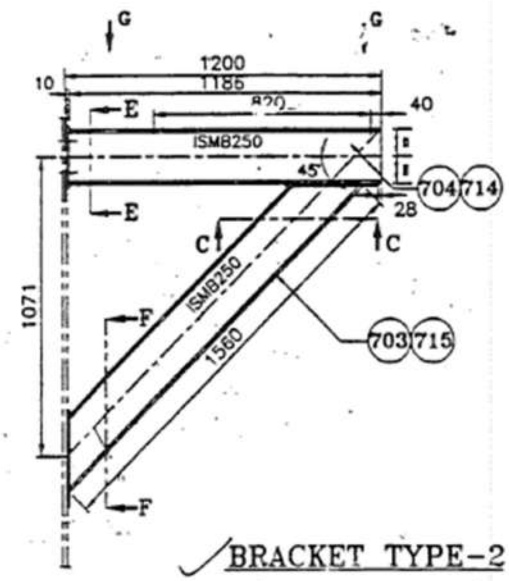
DATE OF INSP. :



BRACING TYPE- 1 TO 6.
406
407
411
VIEW 1-1
170
70
100
ALTERNATIVELY PLACE
MATCH MARK ON
TO SUIT ASSY.

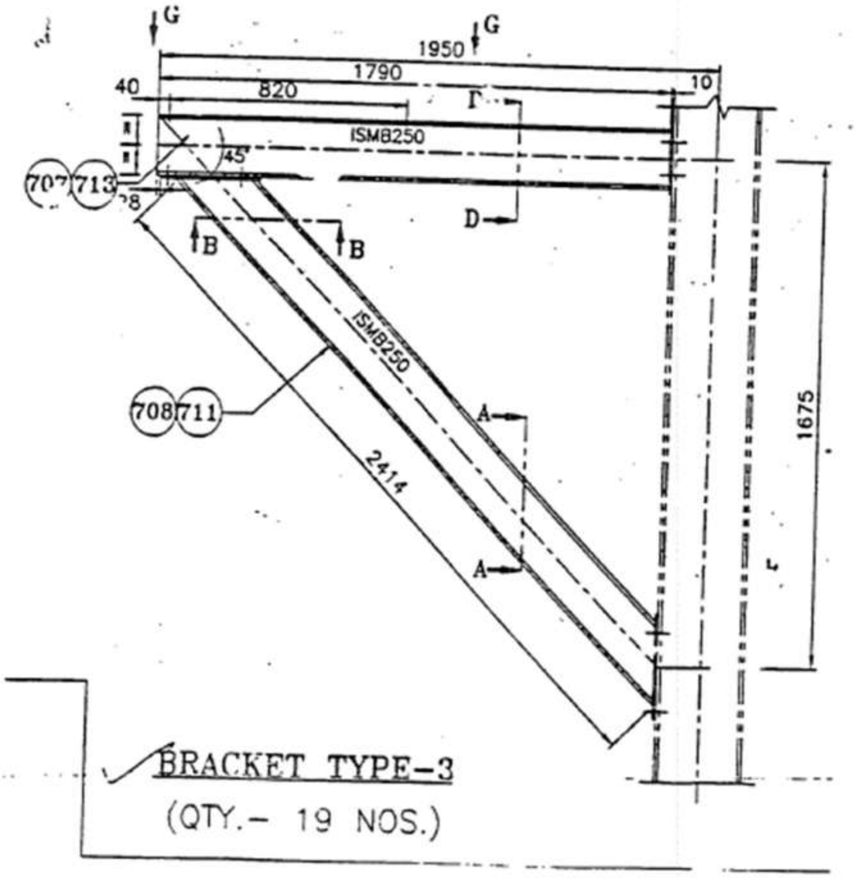


BRACKET TYPE-1 (QTY.- 1 NOS.)



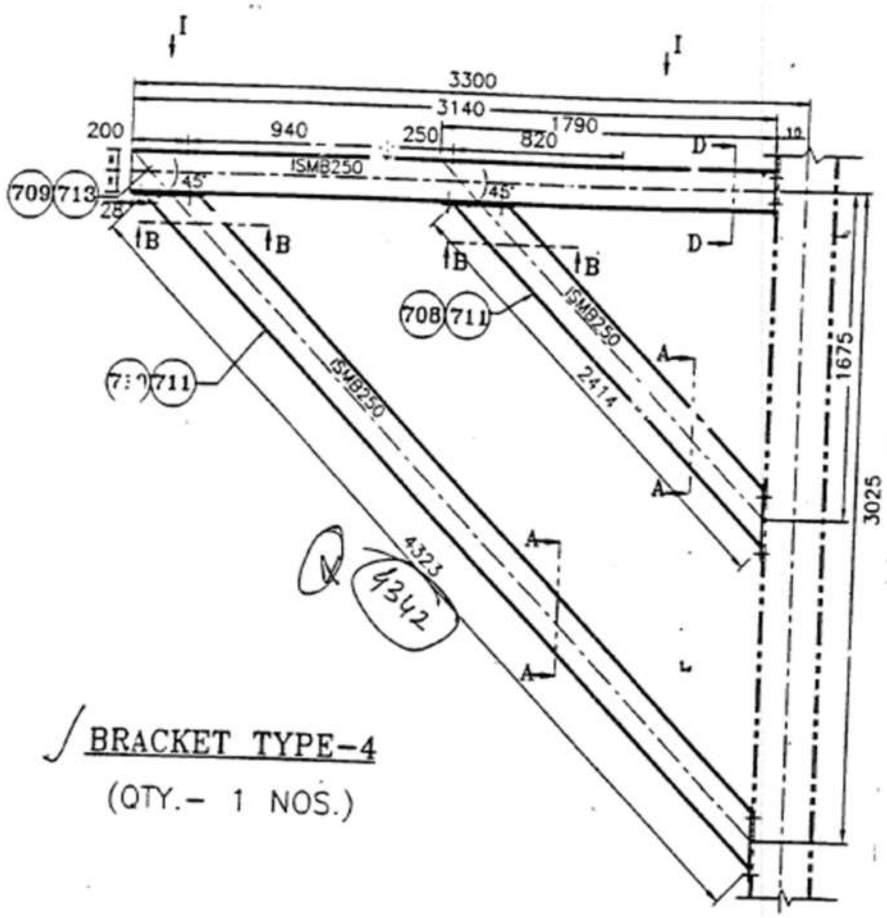
BRACKET TYPE-2

(QTY.– 3 NOS.)



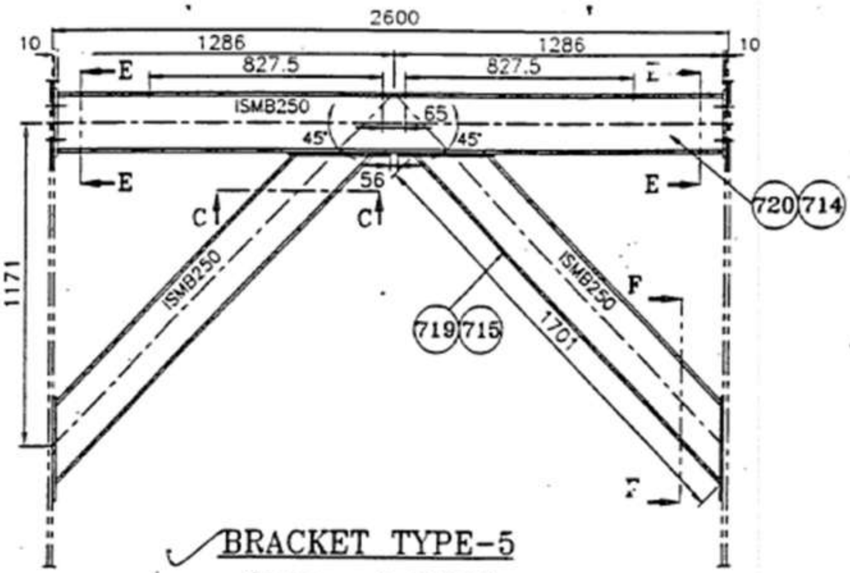
BRACKET TYPE-3

(QTY.- 19 NOS.)

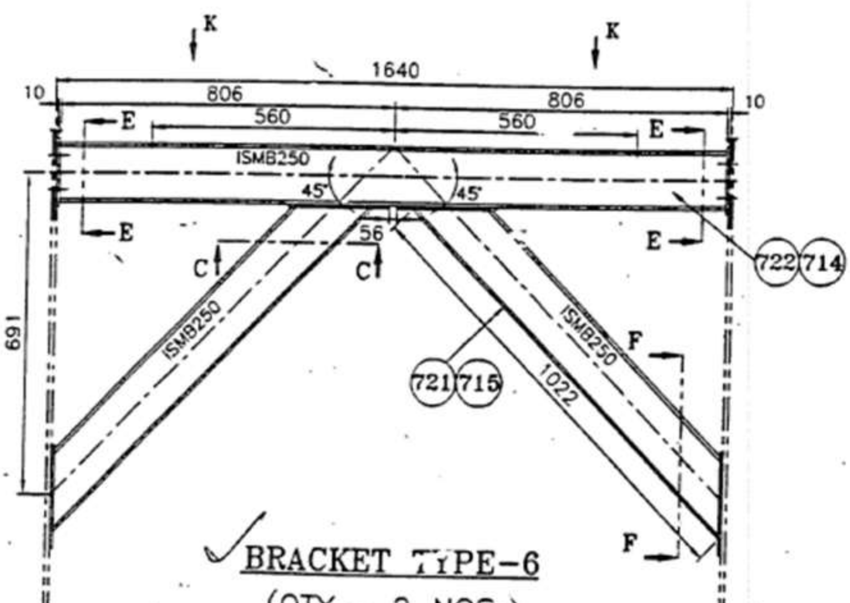


BRACKET TYPE-4

(QTY.- 1 NOS.)

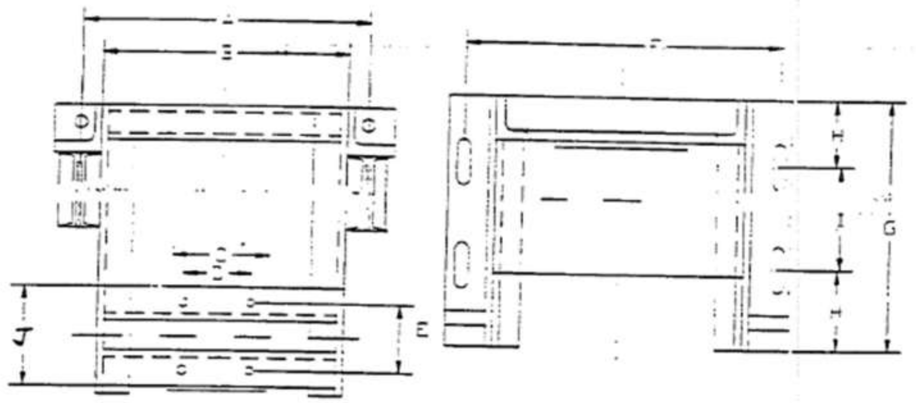


BRACKET TYPE-5
(QTY.- 6 NOS.)



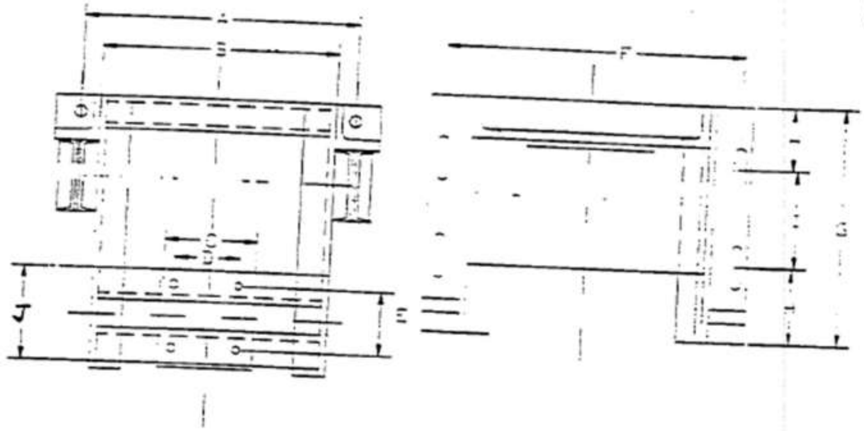
BRACKET TYPE-6
(QTY.- 2 NOS.)

EC 42101



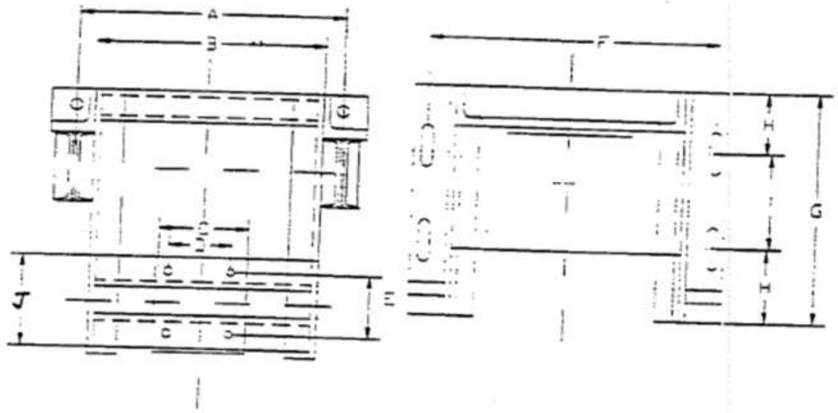
. NO.	DESCRIPTION	DIMENSION OBSERVED IN MM					
		1	2	3	4	5	
1	A = 1300·0	1298/97	1297/99	1298	1299	1298	
2	B = 1155·0	1150/55	1155/55	1152/53	1150/53	1150/52	
3	C = 380·0	380	380	382	383/82	382	
4	D = 279·0	278	280	279/80	279/80	279/80	
5	E = 279·0	280	280	281/80	278/81	279/80	
6	F = 1364·0	1361/63	1362/65	1361/62	1360/63	1362/64	
7	G = 630·0	630	635/35	634	634/35	632	
8	H = 140·0	140/138	138/140	135/40	137/40	137/40	
9	I = 350·0	352/53	355/55	352	354	354/54	
10.	J =						

PG



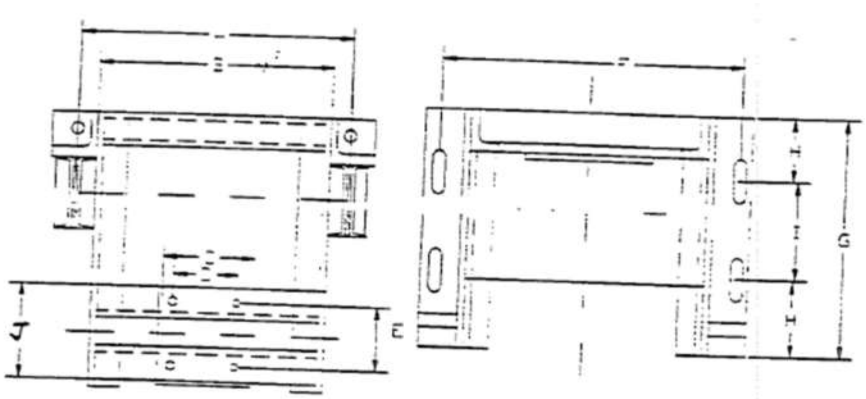
SL NO.	DESCRIPTION	DIMENSION OBSERVED IN MM					
		6	7	8	9	10	
1	A = 1300.0	1298	1299	1298	1298	1295/97	
2	B = 1155.0	1151/52	1157	1152/53	1155/53	1158	
3	C = 380.0	382	381/82	381/82	380/81	381/82	
4	D = 279.0	281	280	281	280	279	
5	E = 279.0	279	280	281	278	279	
6	F = 1364.0	1363/61	1362/64	1366/62	1363/64	1362/62	
7	G = 630.0	632	631	634	634	634	
8	H = 140.0	140/38	140/37	141/37	142/38	142/38	
9	I = 350.0	352	353/50	354	351/52	352/53	
10.	J =						

PG



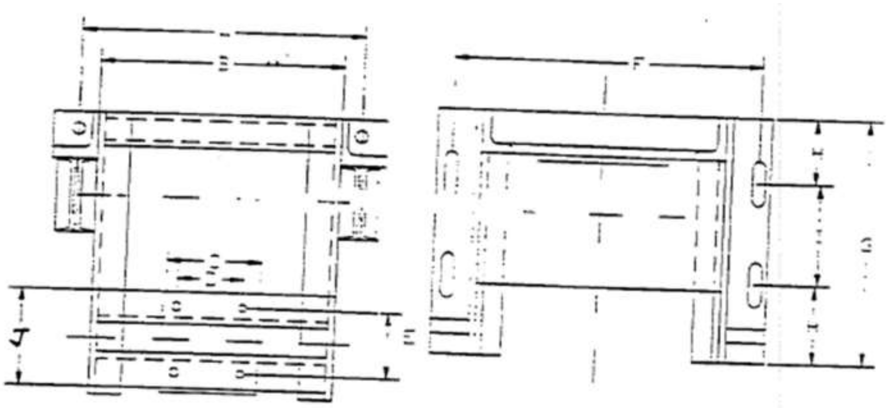
SL NO.	DESCRIPTION	DIMENSION OBSERVED IN MM					F
		11	12	13	14	15	
1	A = 1300.0	1299	1298	1299	1300/01	1299	
2	B = 1155.0	1157/55	1152/51	1154/53	1151	1152	
3	C = 380.0	380	380	382	381	382	
4	D = 279.0	278	278	278	280	281	
5	E = 279.0	280	278	279	281	281	
6	F = 1364.0	1364/64	1362/64	1361/64	1363/61	1363/61	
7	G = 630.0	635	632	634	635	634	
8	H = 140.0	142/38	140/38	140/38	140/38	140/38	
9	I = 350.0	350/51	353/52	354/53	353/53	354/54	
10.	J =						

EC 42103



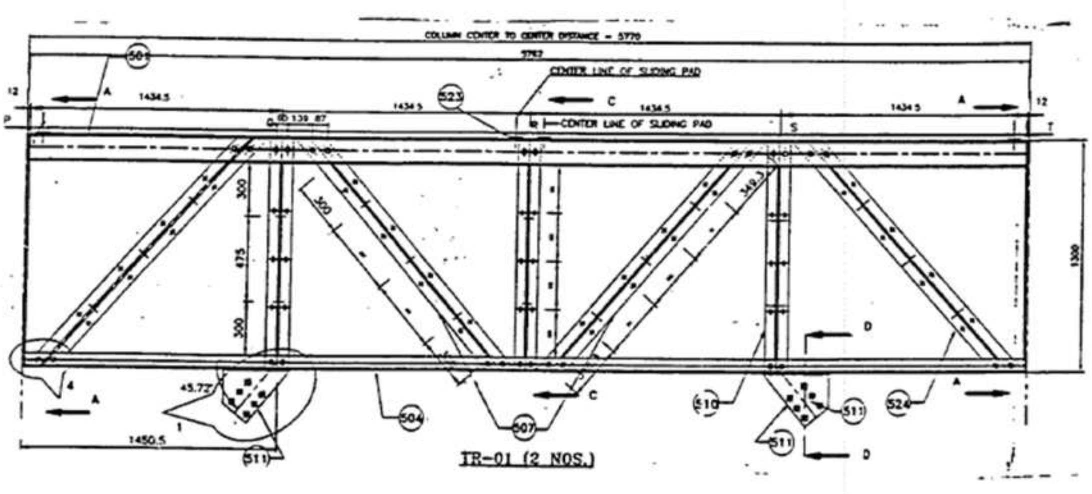
SL NO.	DESCRIPTION	DIMENSION OBSERVED IN MM					
		16	17	18		=	
1	A = 1300.0	1295	1298	1299			
2	B = 1155.0	1152	1157	1156			
3	C = 380.0	380	381	382			
4	D = 279.0	279	278	279			
5	E = 279.0	281	279	281			
6	F = 1364.0	1362/63	1366/62	1361/62			
7	G = 630.0	630/35	636/32	635			
8	H = 140.0	141/37	141/38	141/38			
9	I = 350.0	352/53	352/55	351/52			
10.	J =						

EC 42105

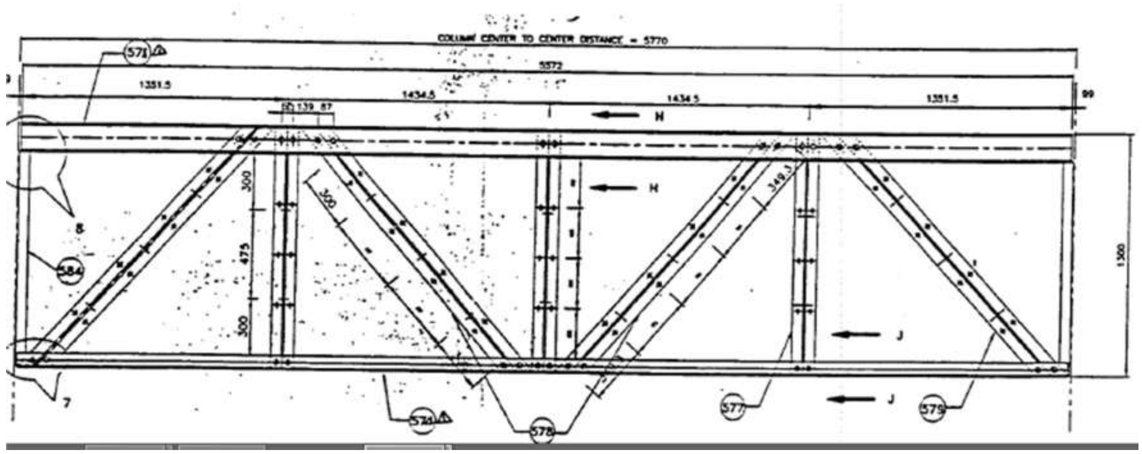


SL NO.	DESCRIPTION	DIMENSION OBSERVED IN MM				
		1	2	3	4	5
1	A = 1300.0	1295/98	1295/98	1299	1298/99	
2	B = 1155.0	1152	1153	1155	1154	
3	C = 400.0	402	401	402	400/401	
4	D = 318.0	318/19	320/20	319/19	320/19	
5	E = 305.0	305/06	305/06	306/05	305/07	
6	F = 1364.0	1365/63	1364/62	1362/61	1363/63	
7	G = 630.0	635/32	632	632/33	632/32	
8	H = 140.0	140/38	140/37	147/38	142/38	
9	I = 350.0	349/48	351/52	352/51	352/51	
10.	J =					

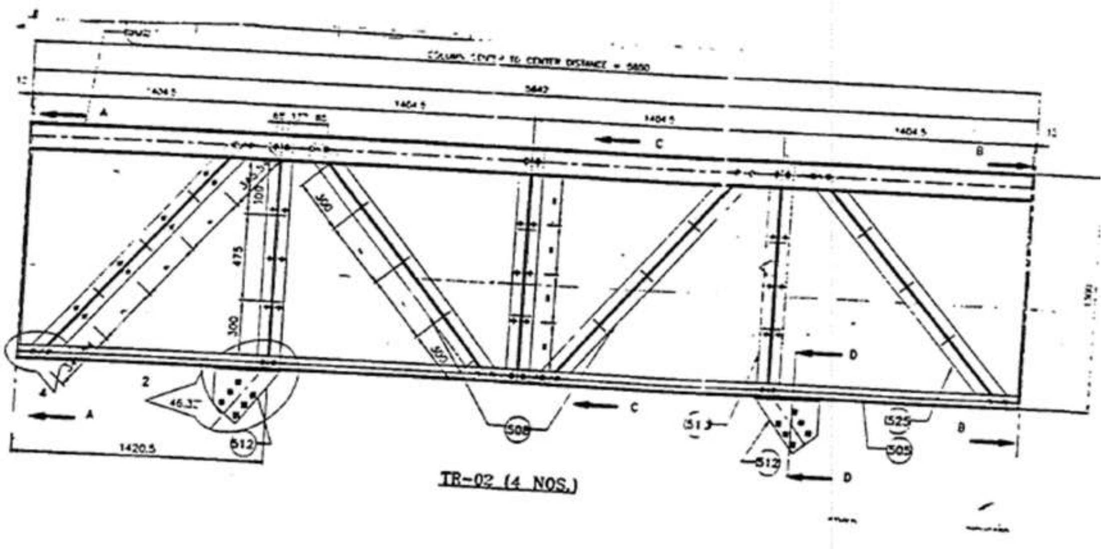
TRUSS ASSEMBLY (TR-01)



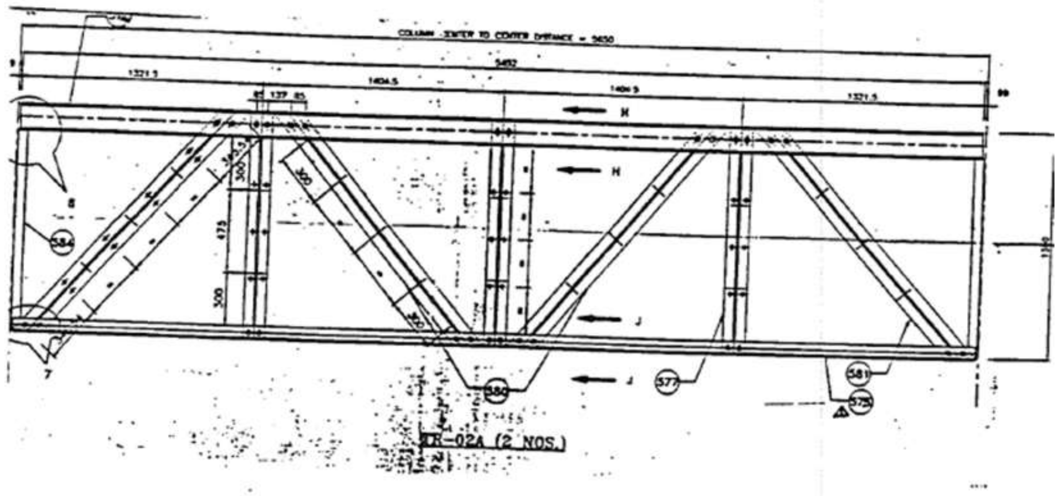
TRUSS ASSEMBLY (TR-01A)



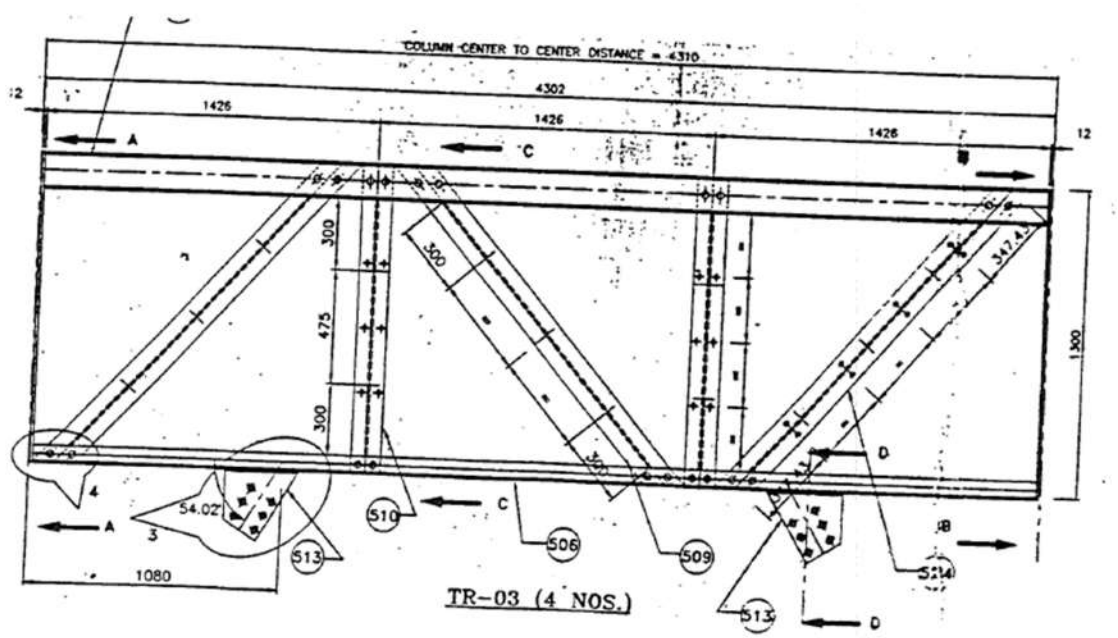
TRUSS ASSEMBLY (TR-02)

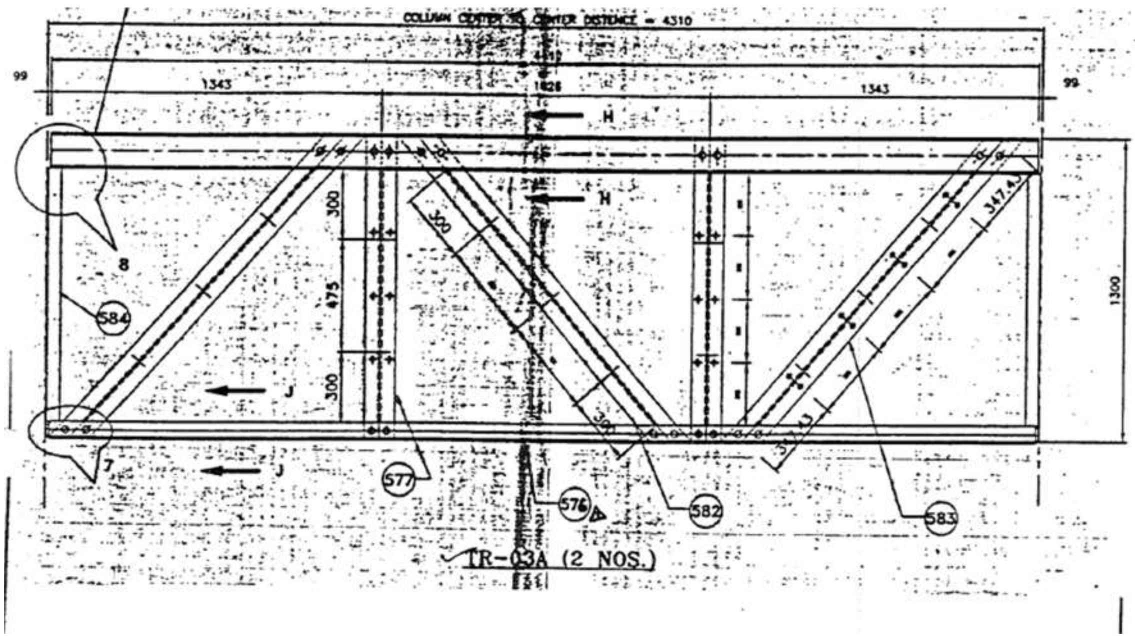


TRUSS ASSEMBLY (TR-02A)

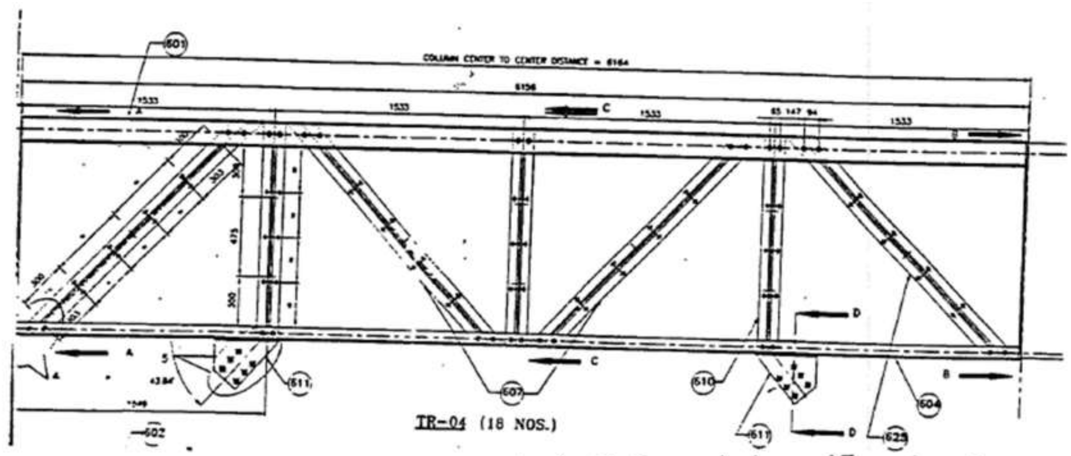


) TRUSS ASSEMBLY (TR-03)

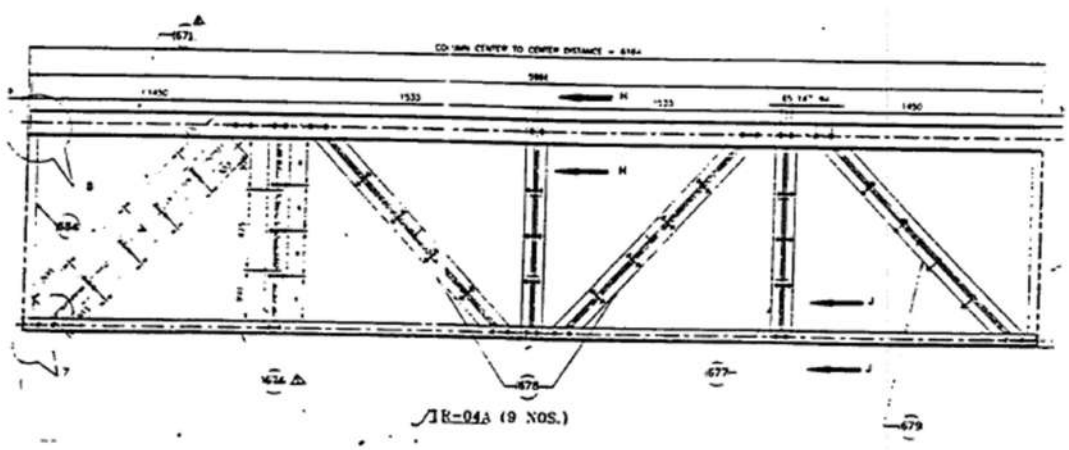




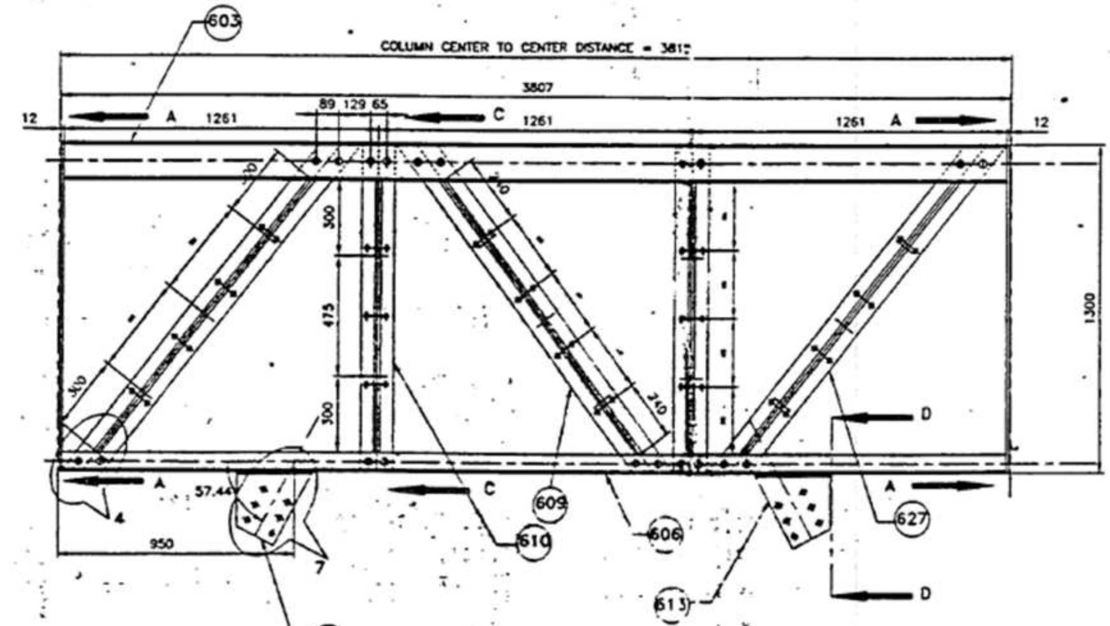
TR-03A (2 NOS.)



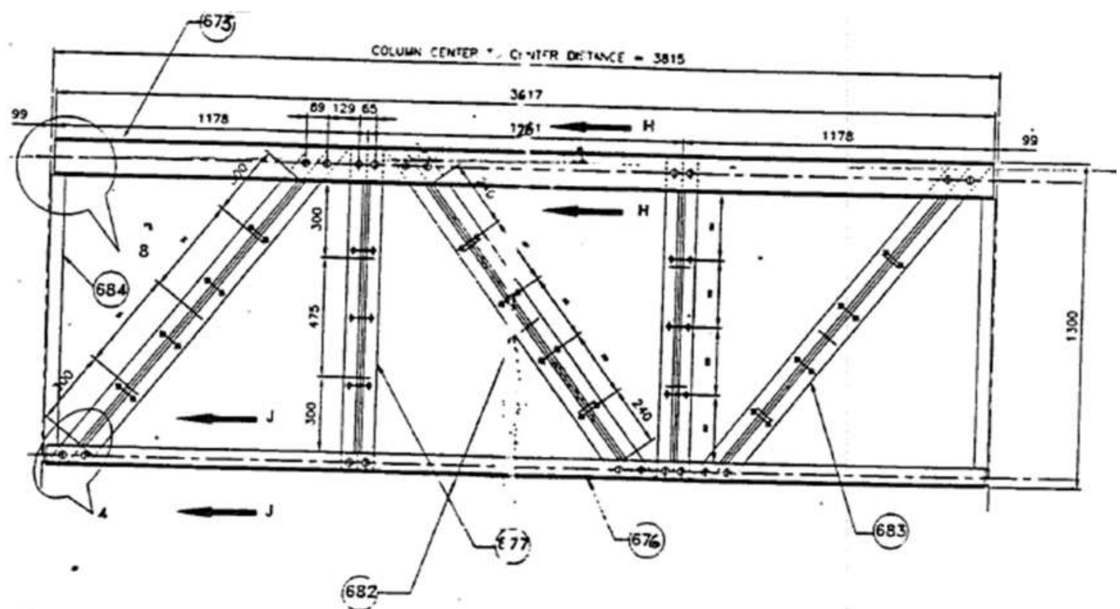
TR-04 (18 NOS.)



TR-04A (9 NOS.)

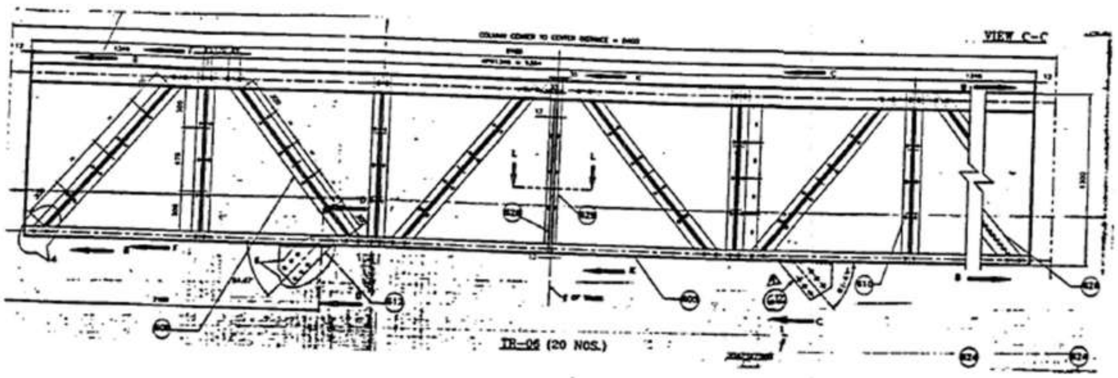


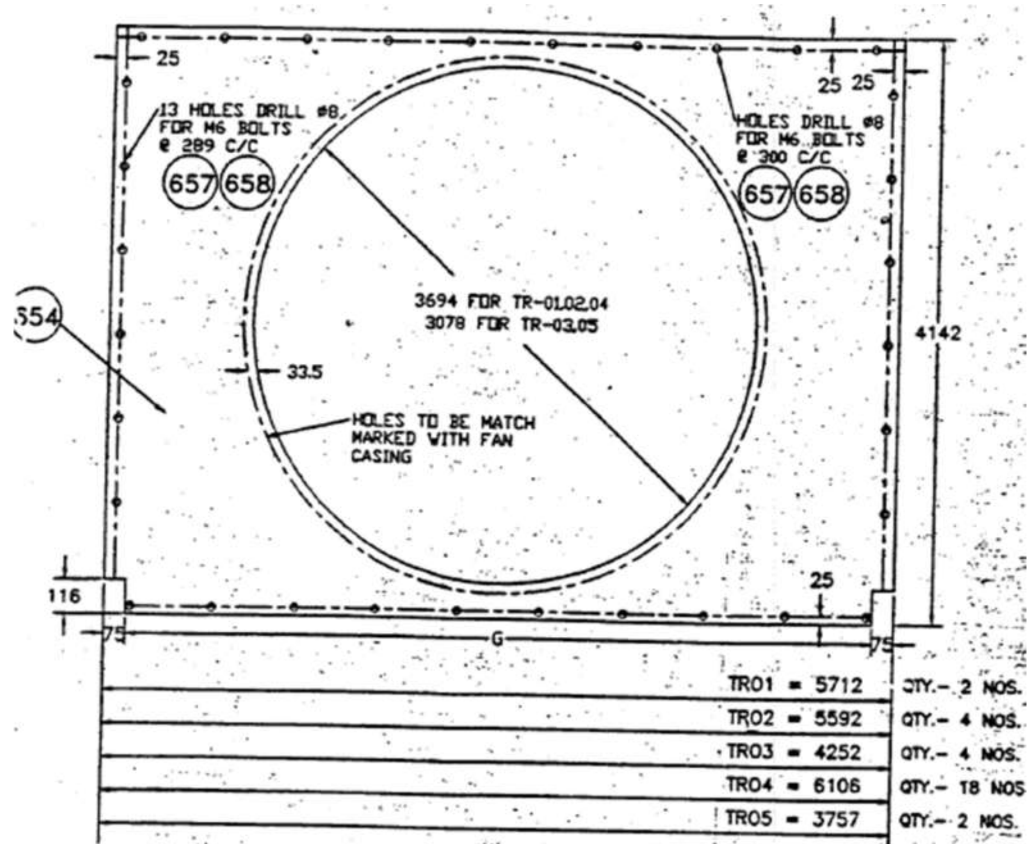
TR-05 (2 NOS.)



TR-05A (1 NOS.)

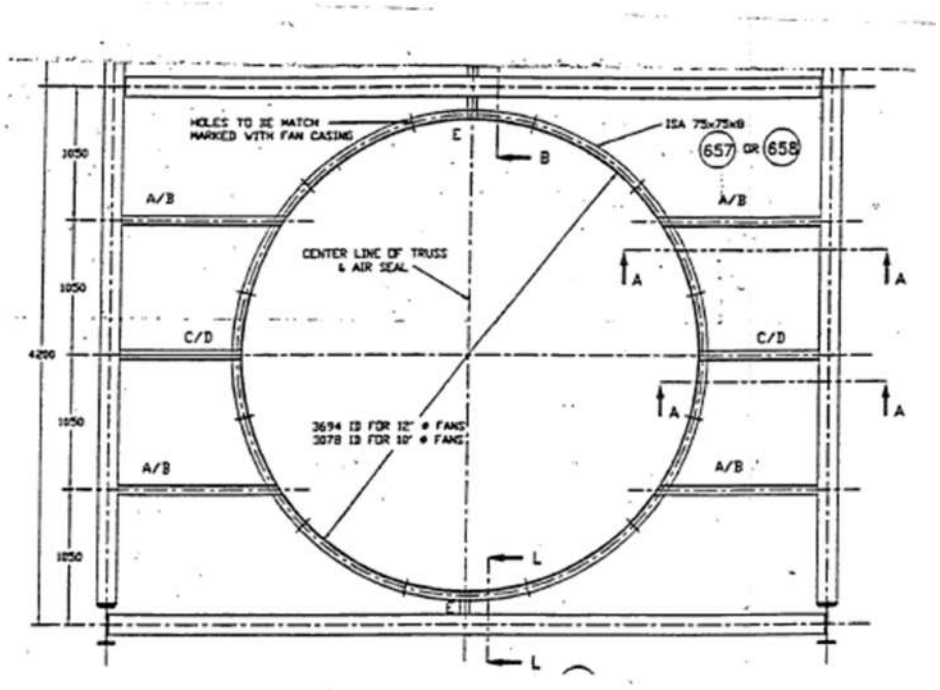
TR-06

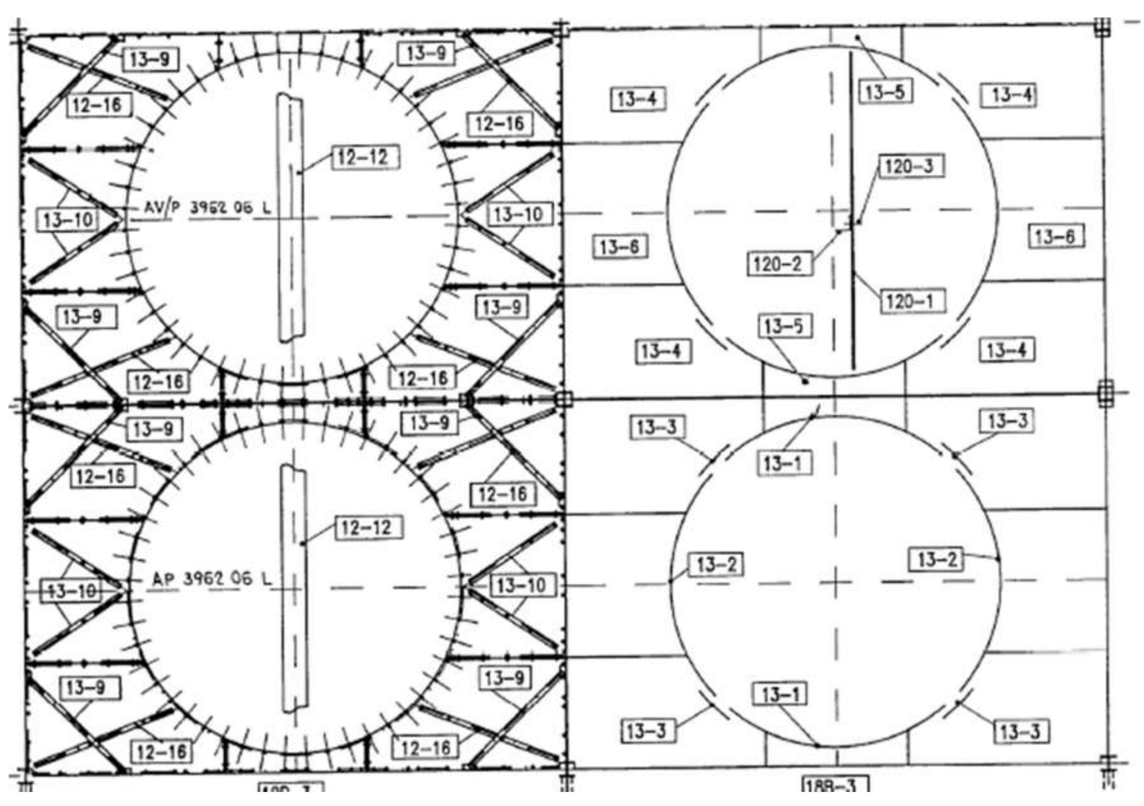




BOTTOM COVER PLATE

TRUSS ARRANGEMENT

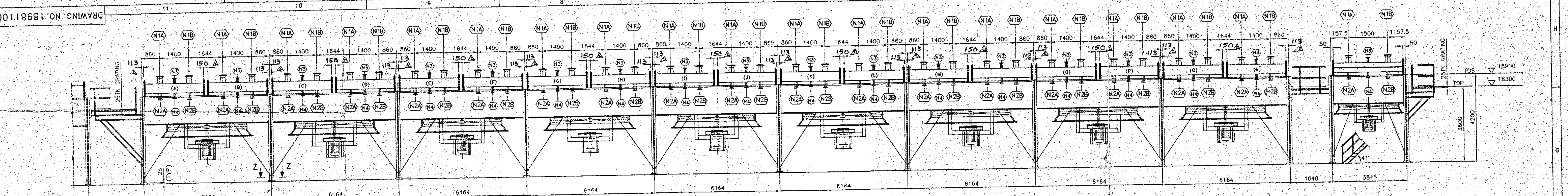




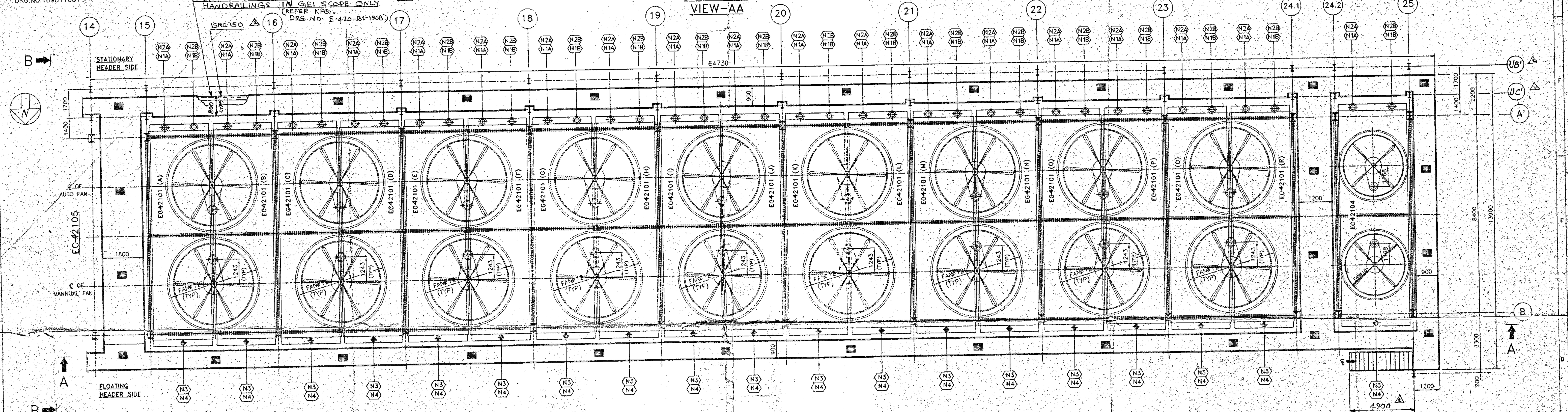
FIRST ANGLE PROJECTION

ALL DIMENSIONS ARE IN MM.

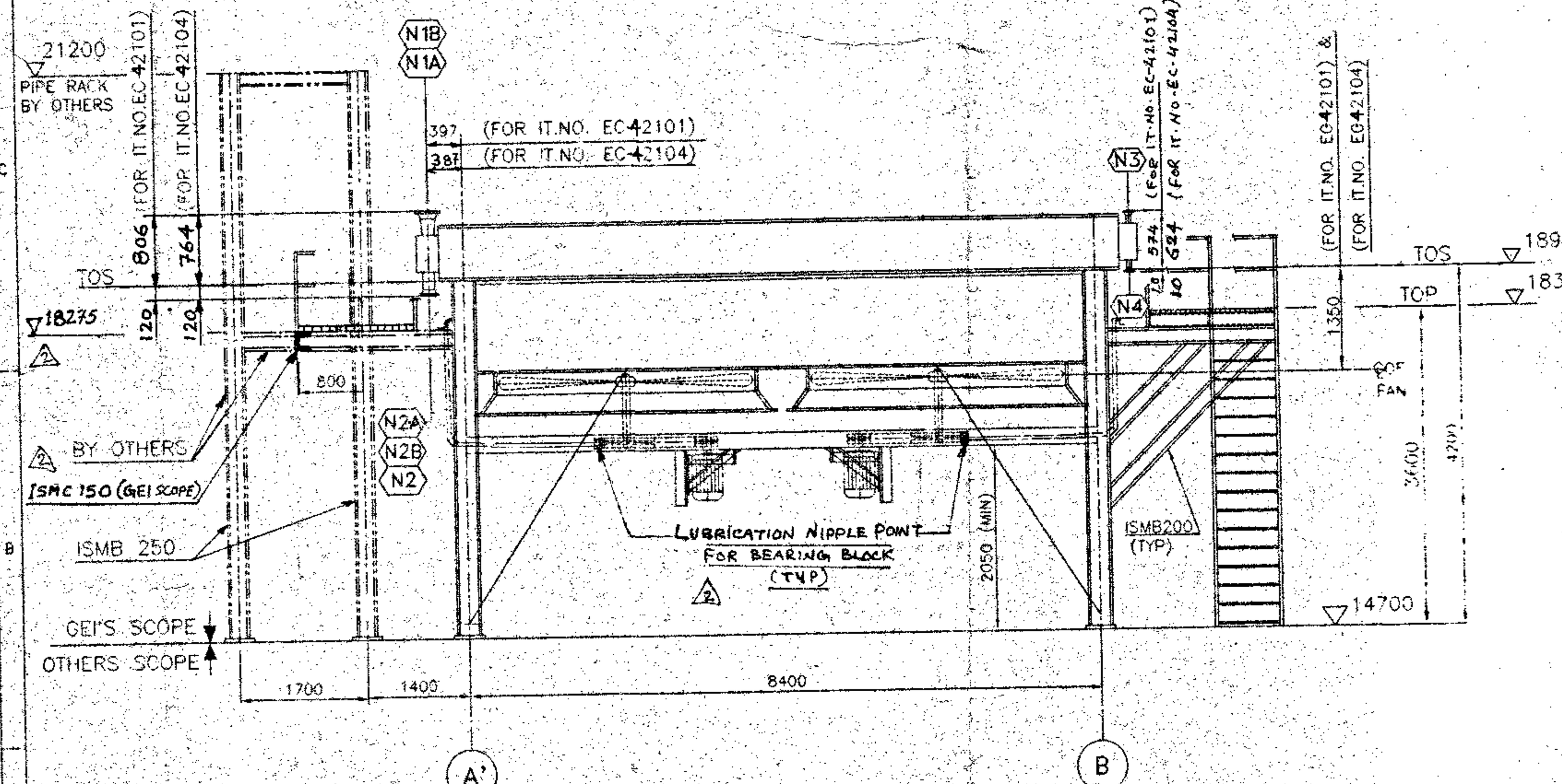
DRAWING NO. 189811000



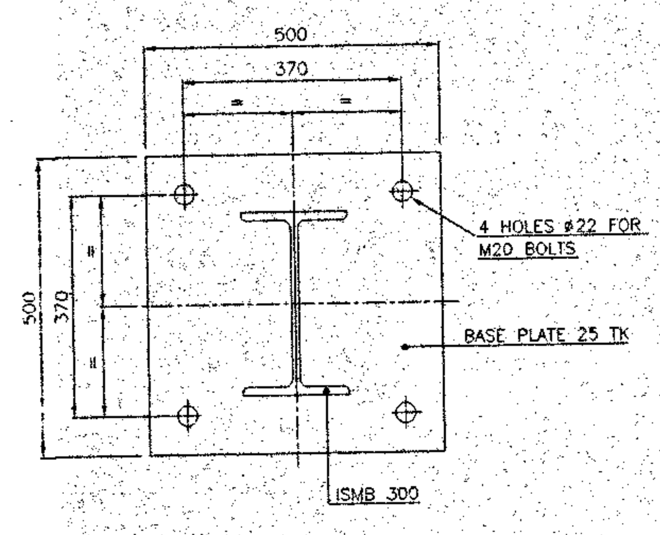
ELEVATION VIEW-AA



PLAN VIEW



VIEW-BB



VIEW-ZZ (TYP)

FAN DATA					MOTOR		
ITEM NO.	FAN Ø	FAN MODEL	MAKE	TYPE (BLADE PITCH ADJUSTMENT)	KW	MAKE	QTY
EC42101 A TO R	Ø12'	SR 40-12'-6	MOORE	9 MANNUAL + 9 AUTO	18.5	CROMPTON GREAVES	18
EC42104	Ø10'	SR 33-10'-4	MOORE	1 MANNUAL + 1 AUTO	15	CROMPTON GREAVES	2

1. VIBRATION SWITCH - FRANK MURPHY (ONE FOR EACH FAN)

NOZZLE DATA									
ITEM NO	NOZZLE MARK	SIZE NB	CLASS	FLANGE	SCH.	MATERIAL PIPE	MATERIAL FLANGE	SERVICE	REMARKS
EC42101 A TO R	N1 A B	8"	150#	WNRF	120	SA-106 GR B	SA-105	INLETS	SMOOTH FINISH 125RA TO 250RA
EC42101 A TO R	N2 A B	6"	150#	WNRF	120	SA-106 GR B	SA-105	OUTLETS	SMOOTH FINISH 125RA TO 250RA
EC42101 A TO R	N3	1 1/2"	150#	WNRF	160	SA-106 GR B	SA-105	VENT+B.F	SMOOTH FINISH 125RA TO 250RA
EC42101 A TO R	N4	1 1/2"	150#	WNRF	160	SA-106 GR B	SA-105	DRAIN+B.F	SMOOTH FINISH 125RA TO 250RA
EC42104	N1 A,B	8"	150#	WNRF	120	SA-106 GR B	SA-105	INLETS	SMOOTH FINISH 125RA TO 250RA
EC42104	N2 A,B	6"	150#	WNRF	120	SA-106 GR B	SA-105	OUTLETS	SMOOTH FINISH 125RA TO 250RA
EC42104	N3	1 1/2"	150#	WNRF	XXS	SA-106 GR B	SA-105	VENT+B.F	SMOOTH FINISH 125RA TO 250RA
EC42104	N4	1 1/2"	150#	WNRF	XXS	SA-106 GR B	SA-105	DRAIN+B.F	SMOOTH FINISH 125RA TO 250RA

NOTES :-

1. ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE SPECIFIED.
2. THIS UNIT HAS BEEN DESIGNED FOR ADDITIONAL EXTERNAL PIPING LOADS, AS PER API-661.
3. SURFACE PREPARATION:- SAND BLASTING TO SA 2 1/2 STANDARD (FOR HEADER & PLENUM)
   SAND BLASTING TO SA 2 STANDARD (FOR STRUCTURE)
4. PAINTING:-
   (a). HEADER :- PRIME COAT:- ONE COAT OF INORGANIC PAINT (HIGH BUILD TYPE) 65 μ /COAT TKN.
   INTERMEDIATE & FINAL COAT :- ONE COAT OF SILICON RESIN HEAT RESISTANT PAINT (DESIGNATED COLOUR) 20 μ /COAT TKN. } AT SITE BY CONTRACTOR
   (b). STRUCTURES & STAIR CASE:- PRIME COAT:- ONE COAT OF EPOXY MASTIC (100 μ /COAT TKN)
   INTERMEDIATE & FINAL COAT:- ONE COAT OF ACRYLIC POLYURETHANE(35 μ /COAT TKN } AT SITE BY CONTRACTOR
   (c). PLENUM CHAMBER, FAN CASING, C-FRAMRE & DRIVE ASSY. }- PRIME COAT:- ONE COAT OF EPOXY PAINT (HIGH BUILD TYPE) (60 μ/COAT TKN.)
   INTERMEDIATE & FINAL COAT:- ONE COAT OF SYNTHETIC RUBBER BASED HEAT RESISTING ALUMINIUM PAINT (20 μ/COAT TKN) } AT SITE BY CONTRACTOR
   (d). ALL GRATING SHALL BE HOT DIP GALVANISED.
   (e). ALL HARDWARES TO BE CADMIUM PLATED.
   (f). FINAL COLOUR COAT:- A.C. GREY
5. FOR MOTOR MAINTENANCE ARRGT. REFER DRG. NO. 1898MLA
6. TOTAL WEIGHT OF ASSY (IT. NO 42101 TO 42105) (STRUCTURE, PLENUM, DRIVE ASSY, FAN, MOTOR & GRATING Etc.) } 125635-00 KGS
7. INTERMEDIATE & FINAL COATS OF PAINTING: WHICH ARE INACCESSIBLE AFTER BUNDLE ASSY, i.e. INSIDE OF TUBE BUNDLE PORTIONS, TUBE SHEET EXTERNAL FACE (FIN TUBE SIDE), BOTTOM PLATES AND C-FRAME; MATCHING TOP PLATES OF SPLIT HEADERS SHALL BE DONE BY GMI.

NAME OF CONSULTANT -	KPG / TEIL			LOCATION -	MANGLORE		
PROJECT -	M.R.P.L- PHASE-II			CLIENT -	M.R.P.L		
DRN.	VA	[signature]	3/7/97	GEI Engineering Ltd. Bhopal India			
CHKD.	AD	[signature]	-"-	PHONE : 586922	FAX : 0755-587678		
APPD.	USP	[signature]	4-7-97				
YEAR OF ORDER 1997	NAME	SIGN	DATE	CUST. P.O.NO. LOI NO. E-2014 DTD. 26/05/97	NO.REQD. 1 NO.	SCALE 1=90	

REV.	DATE	DESCRIPTION	BY	CKD.	APPD.
2	19/1/98	CUSTOMERS COMMENTS INCORPORATED AND MARKED REV. 2	[signature]	[signature]	[signature]
1	23/8/97	CUSTOMERS COMMENTS INCORPORATED	[signature]	[signature]	[signature]

TITLE: SETTING PLAN
ITEM NO. EC-42101 (SERVICE:-RECYCLE SPLITTER REFLUX CONDENSER)
ITEM NO. EC-42104 (SERVICE:-ABSORBER FEED COOLER)

DRAWING NO. 189811000 REV. 2

CAD FILE NAME:- SP1898-1

DETAIL ENGG. & DRG. DONE AT GEI Engineering Ltd. 26-A INDUSTRIAL AREA GOVINDPURA BHOPAL-462 023.